

플랫폼 기반 소재 인공지능 소개 및 실습

한국화학연구원
화학데이터기반연구센터
장 승 훈

교육 내용 및 순서

- 인공지능 플랫폼, ChemAI 소개와 활용 방법 (14:00-14:50)

ChemAI: KRICT AI Platform for Data-Driven Material Research

- 무기형광체 데이터셋 수집 및 광학특성 예측 (15:00-15:50)

Optical property dataset of inorganic phosphor

- 전이학습 활용 고분자 유리전이온도 예측 (16:00-16:50)

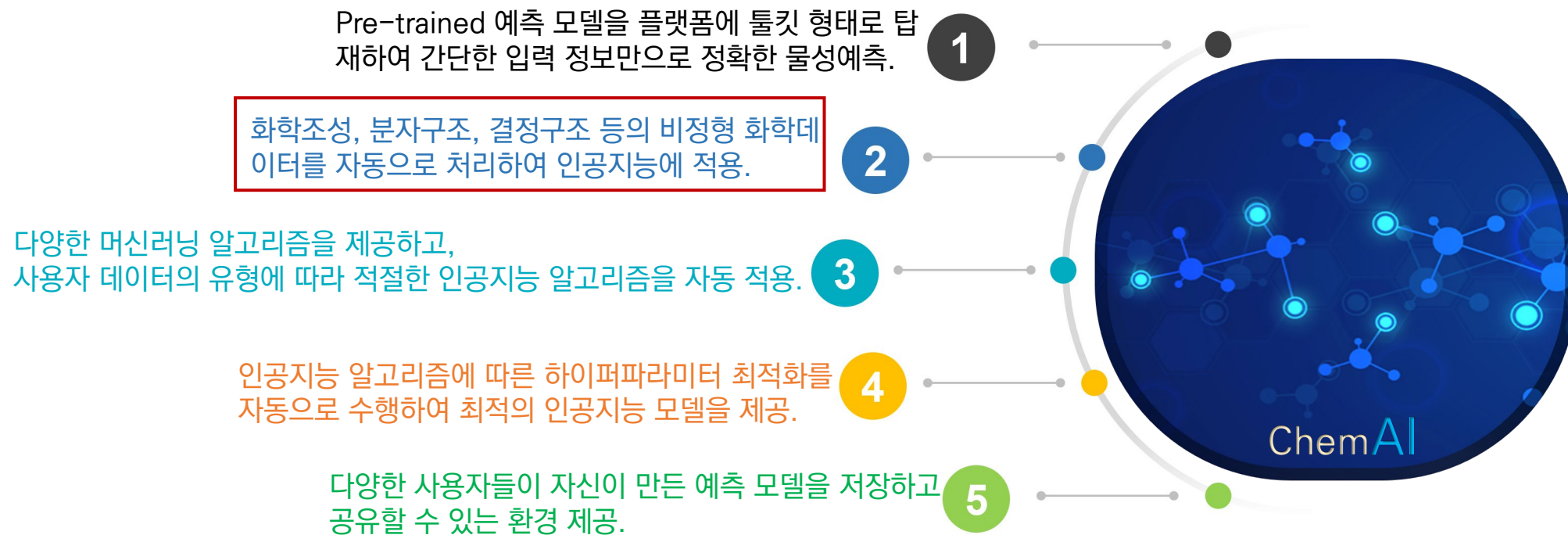
High Glass Transition Temperature Fluorinated Polymers Based on Transfer Learning with Small Experimental Data



인공지능 플랫폼, ChemAI 소개와 활용 방법

ChemAI: KRICT AI Platform for Data-Driven Material Research

화학소재 인공지능 플랫폼: ChemAI



웹기반 환경에서 **컴퓨터 코딩 없이** 사용자가 자신의 데이터를 활용하여 **손쉽게 인공지능 예측 모델을 생성**하고 이를 활용하여 화학 소재 연구를 수행할 수 있는 인공지능 플랫폼

머신러닝 플랫폼 v1, SimPL-ML



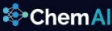
- 누구나 손쉽게 사용자의 데이터를 활용하여 머신러닝 연구를 수행할 수 있는 웹기반 머신러닝 플랫폼 v1 SimPL-ML 개발.
- 사용자 데이터 불러오기 기능.
- 사용자 데이터 편집 기능.
- 화학조성 기반 원자 특성 생성 기능.
- 플랫폼 내 데이터셋 저장 및 관리 기능.
- Example dataset 제공.

- 다양한 ML알고리즘 선택기능.
- 모델 트레이닝 결과 가시화 기능.
- 하이퍼 파라미터 최적화 및 가시화 기능.
- 예측모델 저장 및 관리 기능.

- Pre-trained 화학식 기반 밴드갭 예측 모델 제공.
- 금속/비금속 분류 모델, Binary, Ternary, Quaternary 무기소재 밴드갭 예측모델 탑재.

An Easy, Simple, and Accessible Web-based Machine Learning Platform, SimPL-ML Integrating Materials and Manufacturing Innovation 11, 85–94 (2022)
Seunghun Jang*, Gyoung S. Na, Jungho Lee, Jung Ho Shin, Hyun Woo Kim & Hyunju Chang

화학소재 인공지능 플랫폼: ChemAI



Search...

Seunghun Jang

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Links

ChemAI: KRICT AI Platform for Data-Driven Chemical Sciences

An end-to-end machine learning platform for data-driven chemistry and materials science with user-defined problems

Get examples

- 4개 주요 메뉴로 구성
- 관련 논문 1편 게재
- 특허 2편 등록, 1편 출원
- 기술 이전 1건

Toolkits: Pre-Trained Machine Learning Models

Predict target variables of your chemical data using our pre-trained prediction models. ChemAI provides useful toolkits for machine learning prediction and data preprocessing. All toolkits in the following table is publicly available, and implementation details of the toolkits are given in the [Documentation page](#).

Toolkit	Input Type	Target Property (Unit)	R ² Score
 Band Gap Prediction Prediction of experimental band gap [more]	• Composition	Experimental band gap (eV)	0.909
 Formation Energy Prediction Prediction of experimental formation energy [more]	• Composition	Experimental formation energy (eV/atom)	0.907
 Thermoelectricity Prediction Prediction of ZT (figure of merit) for a given temperature [more]	• Composition • Temperature (K)	ZT (Figure of merit)	0.867
 Band Gap Prediction of Perovskites Band gap prediction of hybrid perovskites [more]	• Crystal structure (.cif)	HSE band gap (eV)	0.901
 Band Gap Correction Prediction of G ₀ W ₀ band gap from naive GGA band gap [more]	• Crystal structure (.cif) • GGA band gap (eV)	G ₀ W ₀ band gap (eV)	0.951

ChemAI – Toolkits

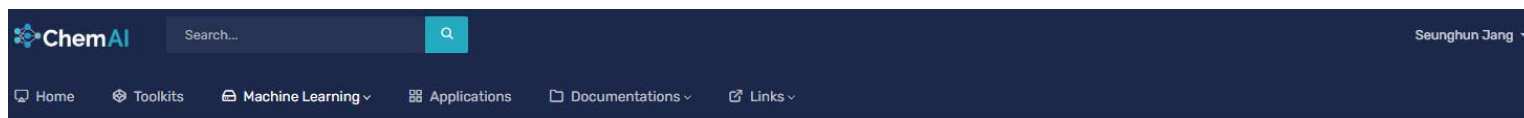
 Band Gap Prediction Prediction of experimental band gap [more]	• Composition	Experimental band gap (eV)	0.909
 Formation Energy Prediction Prediction of experimental formation energy [more]	• Composition	Experimental formation energy (eV/atom)	0.907
 Thermoelectricity Prediction Prediction of ZT (figure of merit) for a given temperature [more]	• Composition • Temperature (K)	ZT (Figure of merit)	0.867
 Band Gap Prediction of Perovskites Band gap prediction of hybrid perovskites [more]	• Crystal structure (.cif)	HSE band gap (eV)	0.901
 Band Gap Correction Prediction of G_0W_0 band gap from naive GGA band gap [more]	• Crystal structure (.cif) • GGA band gap (eV)	G_0W_0 band gap (eV)	0.951
 Prediction of Absorption Max Predicting absorption max of organic chromophores [more]	• Chromophore structure (SMILES) • Solvent structure (SMILES)	Absorption max (nm)	0.902
 Functional Group Detection from IR Spectrum Data Detecting ethanol, butanone, naphthalene, and biphenyl from IR spectrum [more]	• IR spectrum data (.json)	Detection results	Unavailable
 Data Clustering Grouping data points based on density of data [more]	• Data points (.xlsx) • Distance threshold (epsilon) • Quantity threshold (MinPts)	None	Unavailable
 Outlier Detection Detection of abnormal data based on data distribution [more]	• Data points (.xlsx) • Number of neighbors (NumNN)	None	Unavailable
 Maximum PL emission wavelength Prediction Prediction of Maximum PL emission wavelength for inorganic phosphor [more]	• Composition • Temperature • Wavelength of excitation source	Maximum PL emission wavelength (nm)	0.843
 PL decay time Prediction Prediction of PL decay time for inorganic phosphor [more]	• Composition • Temperature • Wavelength of excitation source • PLE detection energy	PL decay time (ns)	0.949
 Configurational stability prediction Binary classification of structural anomaly of surface-adsorbate geometry after DFT calculation [more]	• Surface + Adsorbate geometry (.cif, .vasp, POSCAR)	Configurational stability (structural anomaly after DFT calculations)	Unavailable
 Glass transition temperature prediction Glass transition temperature prediction of copolymer	• Monomer structure (SMILES)	Temperature (K)	
 PCE prediction of OPV Power conversion efficiency prediction of Organic Photovoltaic	• Donor structure (SMILES) • Donor ratio • Acceptor structure (SMILES) • Acceptor ratio • Solvent structure (SMILES)	Power conversion efficiency (%)	0.85
 CIE coordinate prediction CIE coordinate prediction of inorganic phosphor	• Composition • Temperature (K) • Wavelength of excitation source (nm)	CIE x coordinate, CIE y coordinate	0.91 (x), 0.83 (y)

로그인 없이 자유롭게
이용 가능회원가입 및 로그인,
관리자 인증 필요로그인 없이 자유롭게
이용 가능

- pre-trained 예측 모델 개발을 통해 지속적으로 서비스 추가되었음.
- 15종의 툴킷 서비스 제공.

ChemAI – Machine Learning

Model Training



Model Training

Build a machine learning model using your own dataset to predict molecular or materials properties. Implementation details of the machine learning algorithms are presented in the [Documentation page](#). The example datasets are available in Step 1 and their explanations are provided [Datasets page](#).

Step 1: Training Dataset Upload

Select input types of your training dataset. The data handling process for each input type is presented in [Documentation](#).

Example Dataset		qm9_#200_meta.xlsx						Browse
#	Feature Name	Data Type	Mean	Range	Correlation	Input	Target	
1	mol_name	Select data type				☒		
2	smiles	Select data type				☐		
3	dipole_moment	Chemical formula	2.097±1.409	[0.000, 5.498]	-0.398	☐	☐	
4	isotropic_polarizability	SMILES	39.927±9.190	[6.310, 56.800]	-0.072	☒	☐	
5	homo	CIF	-0.263±0.035	[-0.429, -0.181]	-0.498	☒	☐	
6	lumo	Image	0.030±0.053	[-0.111, 0.194]	0.821	☐	☐	
7	band_gap	Numerical	0.293±0.061	[0.155, 0.622]	1.000	☐	☒	
8	electronic_spatial_extent	Numerical	388.364±133.804	[19.000, 721.061]	-0.967	☒	☐	

- Numeric Vector 입력 → Numerical
- Mean, Range, Correlation 정보 제공.
- Chemical Formula, SMILES, Crystal Structure (CIF 파일명), Image (jpg, png 파일명) 입력 → Categorical
- 복수 Input 선택, 단수 Target 선택
- Multi-Modal Learning 기능 탑재.
- Example Dataset 제공 중.

ChemAI – Machine Learning

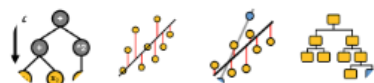
Model Training

Numeric Vector or Chemical formula

Interpretable

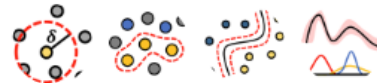
Accurate

Step 2: Model Configuration



- Symbolic Regression
- Linear Regression
- Lasso
- Decision Tree Regression

Step 2: Model Configuration



- K-Nearest Neighbor Regression
- Kernel Ridge Regression
- Support Vector Regression
- Gaussian Process Regression

Step 2: Model Configuration



- Gradient Boosting Tree Regression
- Fully-Connected Neural Network

Smiles or CIF 단일 Input

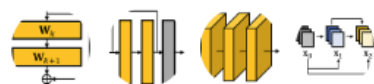
Step 2: Model Configuration



- Crystal Graph Convolutional Neural Network
- Graph Convolutional Network
- Graph Attention Network
- Graph Isomorphism Network
- Unified Message Passing Model
- Edge-Conditioned Filters in Convolutional Neural Networks

Image 단일 input

Step 2: Model Configuration



- Residual Network with 18 Layers
- Residual Network with 34 Layers
- Residual Network with 101 Layers
- Dense Connected Convolutional Networks

이종 input (ex. Image + smiles + numeric vector)

Step 2: Model Configuration

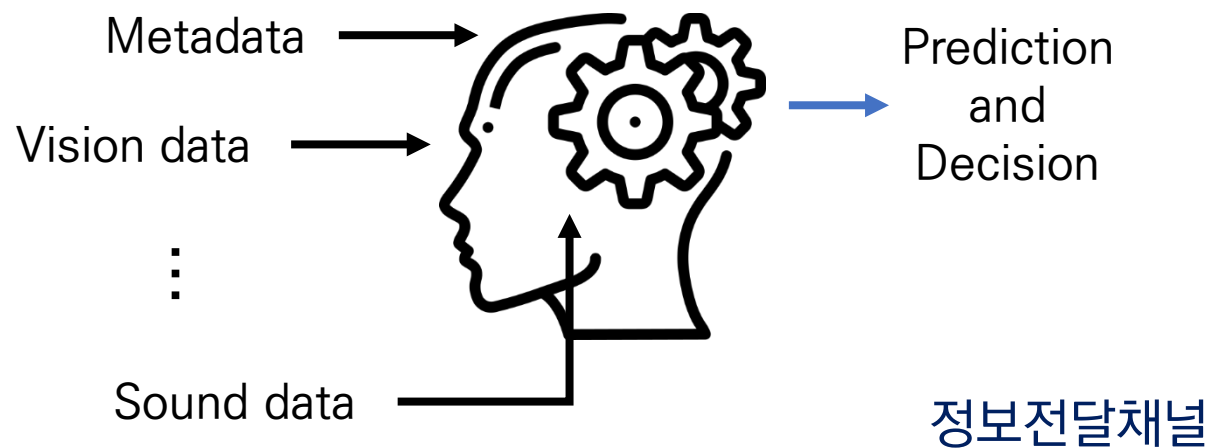


- Multimodal Network

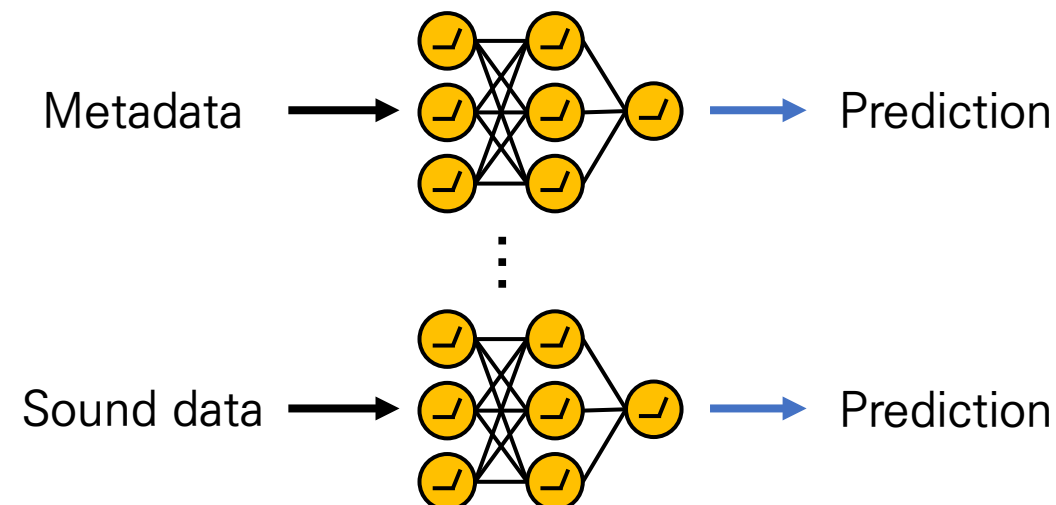
Heterogeneous Data Formats

Multimodal Learning

Human Intelligence for Decision Making



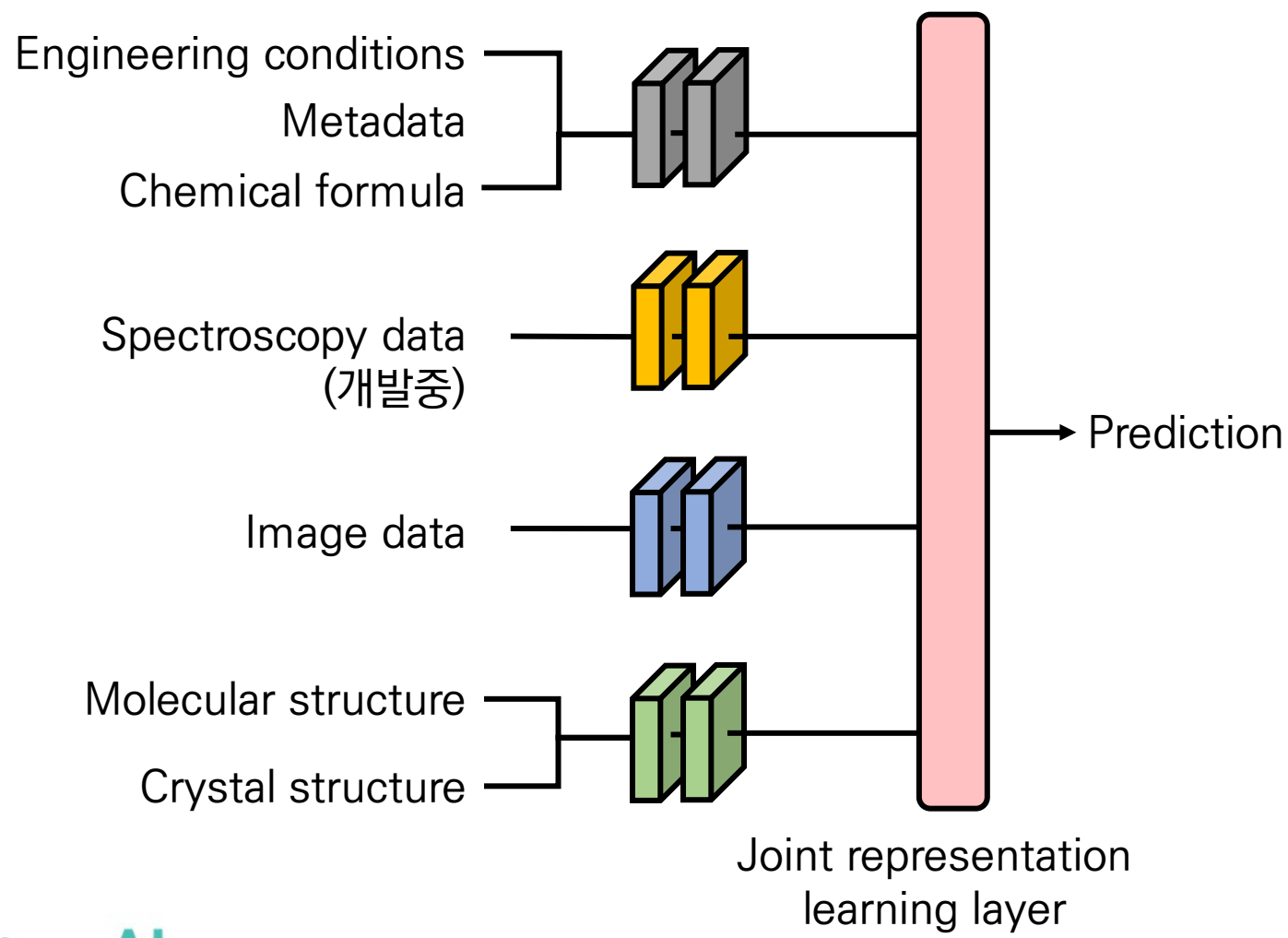
Machine Learning for Heterogeneous Data



Information in the real world usually comes as **different modalities**. Different modalities are characterized by **different statistical properties**. Due to the distinct statistical properties of different information resources, it is **important to discover the relationship between different modalities**. Multimodal learning is an approach to represent the **joint representations** of different modalities.

Andrew Y. Ng. Multimodal Deep Learning. ICML, 2011.

Heterogeneous Data Formats in Chemical Applications



등록특허, 다양한 비정형 화학데이터를 이용한 물질의 물성 예측 방법 및 그 시스템

Multimodal learning 구현에 사용된 핵심 Code

```
for dconfig in config_file['dataset_config']:
    idx_input = [int(d) for d in dconfig['idx_input'].split(',')]

    if dconfig['dtype'] == DTYPE_FVEC:
        dataset = ailca.chem.featvec.load_dataset(path_data_file=dconfig['path_dataset'],
                                                  idx_feat=idx_input,
                                                  idx_target=dconfig['idx_target'],
                                                  verbose=False)

        dataset.normalize()
        datasets.append(dataset)
    elif dconfig['dtype'] == DTYPE_FORM:
        for idx in idx_input:
            dataset = ailca.chem.formula.load_dataset(path_data_file=dconfig['path_dataset'],
                                                      elem_attrs=elem_attrs,
                                                      idx_form=idx,
                                                      idx_target=dconfig['idx_target'],
                                                      verbose=False)

            datasets.append(dataset)
    elif dconfig['dtype'] == DTYPE_MSTRUCT:
        for idx in idx_input:
            dataset = ailca.chem.molecule.load_dataset(path_data_file=dconfig['path_dataset'],
                                                         elem_attrs=elem_attrs,
                                                         idx_smiles=idx,
                                                         idx_target=dconfig['idx_target'],
                                                         verbose=False)

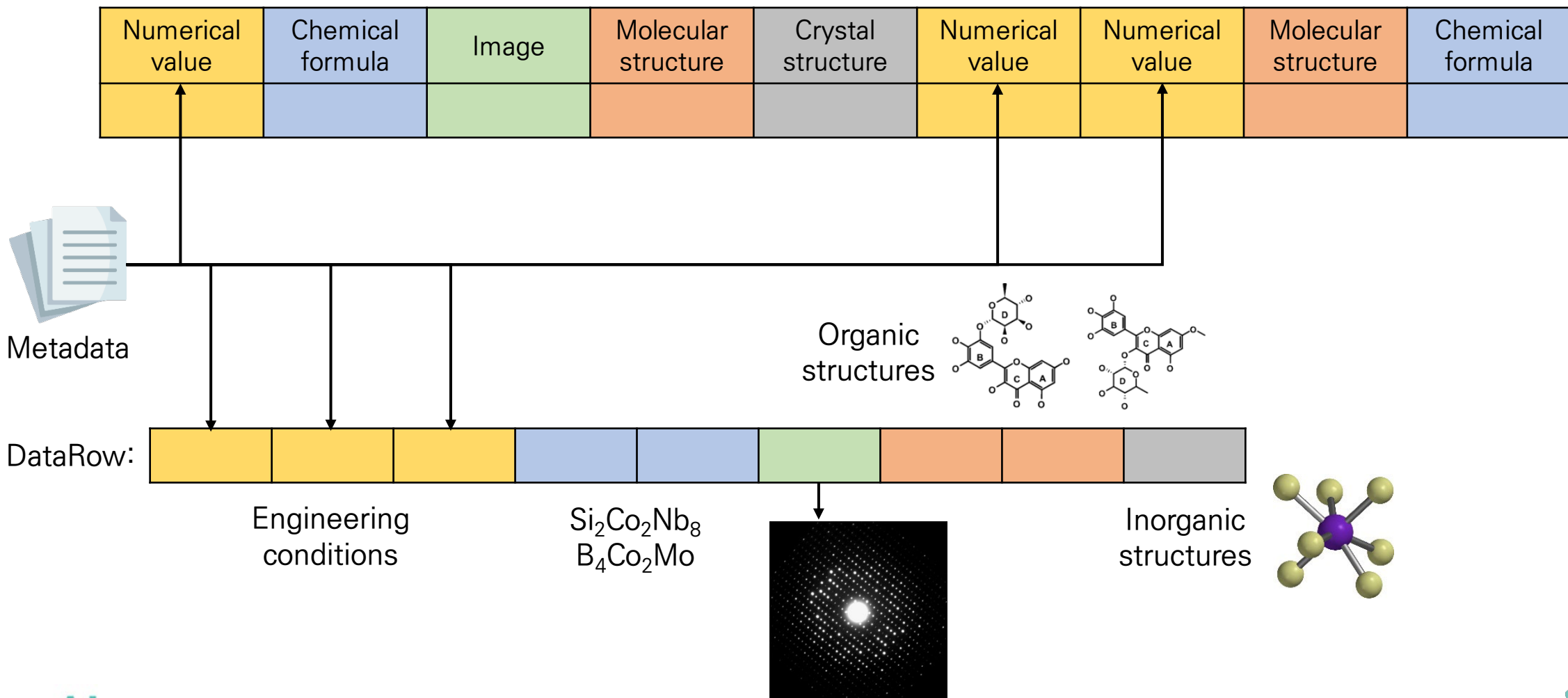
            datasets.append(dataset)
```

```
elif dconfig['dtype'] == DTYPE_CSTRUCT:
    for idx in idx_input:
        dataset = ailca.chem.crystal.load_dataset(path_metadata_file=dconfig['path_metadata_file'],
                                                  path_structs=dconfig['path_structs'],
                                                  elem_attrs=elem_attrs,
                                                  idx_mat_id=idx,
                                                  idx_target=dconfig['idx_target'],
                                                  n_kerl_centers=64,
                                                  verbose=False)

        datasets.append(dataset)
    elif dconfig['dtype'] == DTYPE_IMG:
        for idx in idx_input:
            dataset = ailca.chem.image.load_dataset(path_metadata_file=dconfig['path_metadata_file'],
                                                    path_imgs=dconfig['path_imgs'],
                                                    idx_img_id=idx,
                                                    idx_target=dconfig['idx_target'],
                                                    ext_img=dconfig['ext_img'],
                                                    verbose=False)

            datasets.append(dataset)
```

Multimodal learning에서 data 처리 과정



Multimodal Learning 기반 인공지능 기술 이전

제 C-2022-044797 호



저작권 등록증

1. 저작물의 제호(명칭)	멀티모달러닝 기반 인공지능 모델 생성 기술 및 웹기반 운영 시스템.
2. 저작물의 종류	컴퓨터프로그램저작물>응용프로그램>과학기술
3. 저작자 성명(법인명)	한국화학연구원 대전광역시 유성구 가정로
4. 생년월일(법인등록번호)	160171-0000163
5. 창작연월일	2022년02월01일
6. 공표연월일	2022년10월03일
7. 등록연월일	2022년11월14일
8. 등록사항	저작자 : 한국화학연구원, 창작 : 2022.02.01, 공표 : 2022.10.03

「저작권법」 제53조에 따라 위와 같이 등록되었음을 증명합니다.

2022년 11월 14일

한국저작권위원회

<프로그램 등록, C-2022-044797>

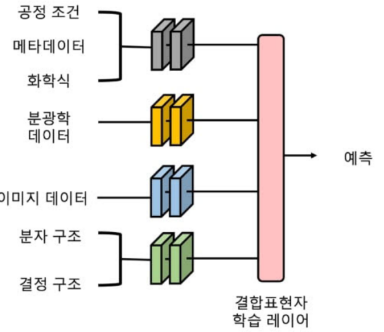
전자신문 etnews 40th Conference allshowTV

경제·금융 전자·모빌리티 통신·미디어·게임 소재·부품 SW·보안 산업·에너지·환경 플랫폼

버추얼랩, 화학연과 '머신러닝 기반 신소재 개발' 기술 이전 계약 체결

발행일 : 2022-12-25 00:00 지면 : 2022-12-26 24면

멀티모달 기반 인공지능 모델 생성 기술



공정 조건, 메타데이터, 화학식, 분광학 데이터, 이미지 데이터, 분자 구조, 결정 구조가 결합표현자 학습 레이어를 거쳐 예측된다.

클라우드 기반 소재 시뮬레이션 플랫폼 전문기업 버추얼랩(대표 이민호)이 한국화학연구원으로부터 '멀티모달 러닝(Multimodal Learning) 기반 인공지능(AI) 모델 생성'에 대한 기술이전을 마쳤다고 25일 밝혔다.

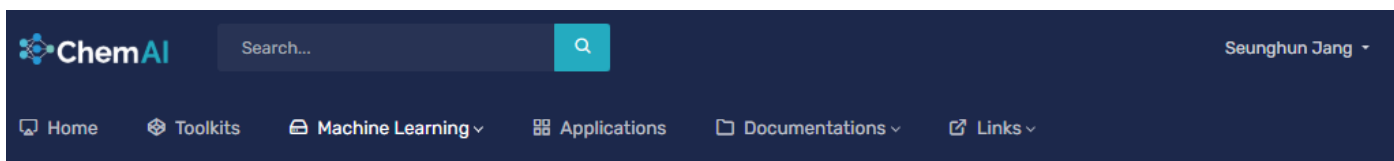
이에 앞서 버추얼랩은 작년 5월 한국과학기술연구원(KIST)과 재료 합성에너지 예측용 머신러닝 알고리즘 'SGCNN(Slab Graph Convolutional Neural Network)' 기술 이전 계약을 맺는 등 머신러닝 기술을 소재 시뮬레이션 플랫폼에 도입하는 노력을 기울이고 있다.

한국화학연구원 나경석 연구원·장승훈·장현주 박사 연구팀이 개발한 '멀티모달 러닝 기반 AI 모델 생성 기술'은 신소재 개발을 위해 재료 분야에서 생성되는 다양한 이중 데이터를 자동으로 처리하는 AI 모델 설계 기능이다.

<멀티모달 러닝 기반 AI 모델 생성에 대한 기술이전 관련 기사> 14

ChemAI – Machine Learning

Training Job



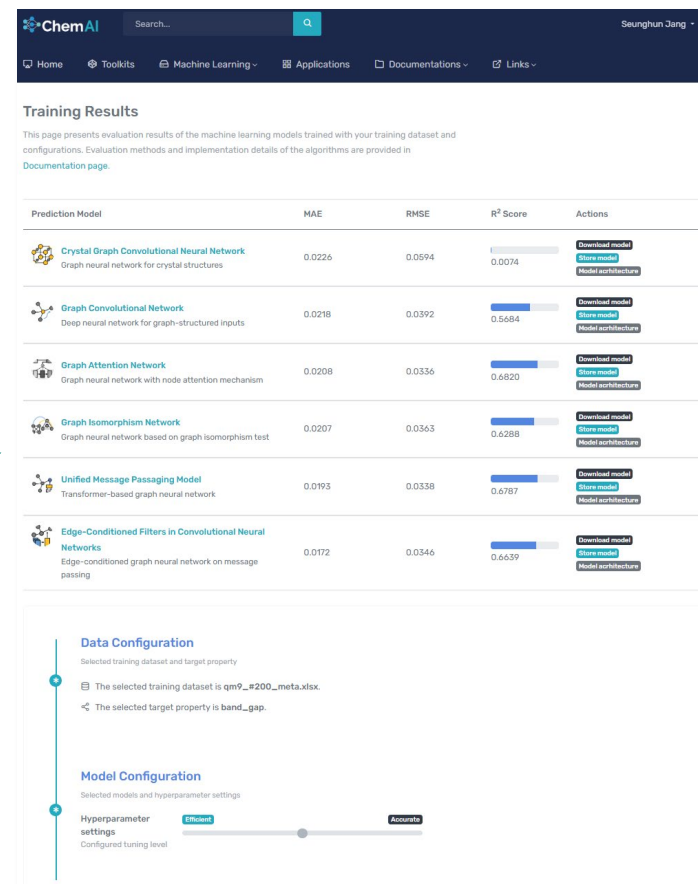
ChemAI > Machine Learning > Training Jobs

Training Jobs

In your account, total 56 training jobs have been completed or are in progress.

Job Id	Job Name	Training Dataset	Status	Started (Running time)	Actions
239	239-THv3	qm9_#200_meta.xlsx	Running	2025-06-27 15:05:10 (18m 23s)	
238	238-THv3	qm9_#200_meta.xlsx	Finished	2025-06-27 15:03:08 (10m 3s)	

훈련 결과 확인을 위해서 Job Name 클릭.



- 사용자별 Training Job 스케줄 관리
- Training Dataset, Status, Running time 정보 표시
- 정상 종료시 사용자 이메일로 안내 메일 자동 송부

ChemAI – Machine Learning

Training Results

ChemAI Search... Seunghun Jang

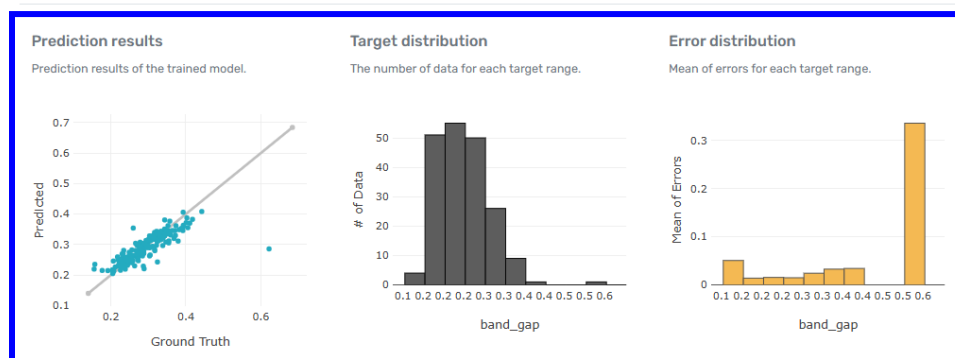
Home Toolkits Machine Learning Applications Documentations Links

Training Results

This page presents evaluation results of the machine learning models trained with your training dataset and configurations. Evaluation methods and implementation details of the algorithms are provided in [Documentation page](#).

Prediction Model	MAE	RMSE	R ² Score	Actions
Crystal Graph Convolutional Neural Network Graph neural network for crystal structures	0.0226	0.0594	0.0074	Download model Store model Model architecture
Graph Convolutional Network Deep neural network for graph-structured inputs	0.0218	0.0392	0.5684	Download model Store model Model architecture
Graph Attention Network Graph neural network with node attention mechanism	0.0208	0.0336	0.6820	Download model Store model Model architecture
Graph Isomorphism Network Graph neural network based on graph isomorphism test	0.0207	0.0363	0.6288	Download model Store model Model architecture
Unified Message Passing Model Transformer-based graph neural network	0.0193	0.0338	0.6787	Download model Store model Model architecture
Edge-Conditioned Filters in Convolutional Neural Networks Edge-conditioned graph neural network on message passing	0.0172	0.0346	0.6639	Download model Store model Model architecture

Prediction Model	MAE	RMSE	R ² Score	Actions
Crystal Graph Convolutional Neural Network Graph neural network for crystal structures	0.0226	0.0594	0.0074	Download model Store model Model architecture
Graph Convolutional Network Deep neural network for graph-structured inputs	0.0218	0.0392	0.5684	Download model Store model Model architecture
Graph Attention Network Graph neural network with node attention mechanism	0.0208	0.0336	0.6820	Download model Store model Model architecture
Graph Isomorphism Network Graph neural network based on graph isomorphism test	0.0207	0.0363	0.6288	Download model Store model Model architecture
Unified Message Passing Model Transformer-based graph neural network	0.0193	0.0338	0.6787	Download model Store model Model architecture

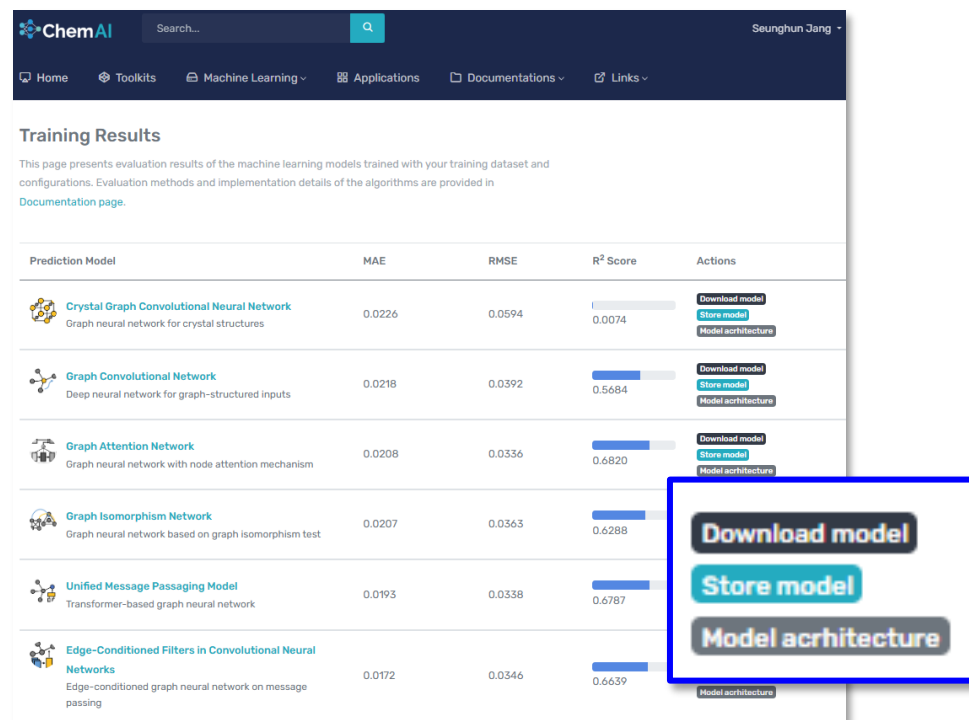


- 6종의 예측 알고리즘 동시 훈련 진행
- MAE, RMSE, R² 모델 훈련 지표 활용 최적 예측모델 선정 (사용자)
- 테스트셋에 대한 Scatter plot 제공
- Target distribution, Error distribution 제공

Edge-Conditioned Filters in Convolutional Neural Networks Edge-conditioned graph neural network on message passing	0.0172	0.0346	0.6639	Download model Store model Model architecture
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ChemAI – Machine Learning

Store Model



Prediction Model	MAE	RMSE	R ² Score	Actions
Crystal Graph Convolutional Neural Network Graph neural network for crystal structures	0.0226	0.0594	0.0074	Download model Store model Model architecture
Graph Convolutional Network Deep neural network for graph-structured inputs	0.0218	0.0392	0.5684	Download model Store model Model architecture
Graph Attention Network Graph neural network with node attention mechanism	0.0208	0.0336	0.6820	Download model Store model Model architecture
Graph Isomorphism Network Graph neural network based on graph isomorphism test	0.0207	0.0363	0.6288	Download model Store model Model architecture
Unified Message Passing Model Transformer-based graph neural network	0.0193	0.0338	0.6787	Download model Store model Model architecture
Edge-Conditioned Filters in Convolutional Neural Networks Edge-conditioned graph neural network on message passing	0.0172	0.0346	0.6639	Download model Store model Model architecture

- 최적 예측모델 오른쪽 “Store model”클릭

1. Metadata of Application

Enter a brief description and summarize metadata of your application.

Title *

Contact E-mail *

Category *

Description

Training Dataset

Input Type *

Target Property *

Unit *

2. Visibility of Application

Select storing options that determine visibility of your application. There are three options for private and public scopes.

☒ Option 1: All in private repository

Your application will be stored in a **private repository** of your account. If the application is stored privately, no one can access this application.

☐ Option 2: Runnable in public repository

Your application with only a runnable model will be stored in **the public repository** of the platform. All users of the platform can access to all information about your model except for the model file. The users can predict their data using your application, but they cannot develop a new application using your prediction model.

☐ Option 3: All in public repository

All information about your application will be publicly opened in **the public repository** of the platform. All users of the platform can publicly use your application in developing another applications. It will be very helpful for users who want to employ your trained model in their applications.

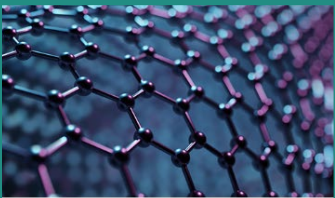
- Metadata 입력
- Visibility 선택
- 모델 저장 클릭
- Repository (개인 사용자 앱 관리)
- Application (플랫폼 전체 앱 관리)

ChemAI – Machine Learning

Repository

Repository

This page lists your applications stored privately or publicly. Currently, total 5 applications have been stored in your repository.



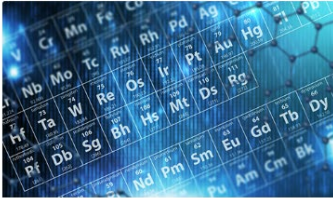
SMILES2Gap prediction

This application predicts **gap (eV)** from molecular structure based on smiles (text and .xlsx).

R² Score: 0.6787

Interpretability

Private Materials Science



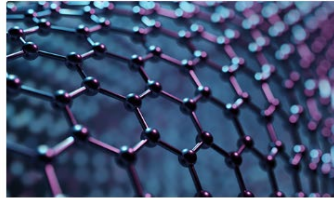
pH prediction

This application predicts **ph (pH)** from numeric vector (text and .xlsx).

R² Score: 0.9239

Interpretability

Private Chemistry



test_model

This application predicts **test (test)** from numeric vector (text and .xlsx).

R² Score: 0.9160

Interpretability

Reusable Runnable Materials Science

저장된 모델 클릭

- 사용자별 예측 모델 관리
- 모델 공유 → Application
- Pre-trained 모델 활용 Prediction시 활용.

모델 로드

SMILES2Gap prediction

This application was implemented to predict **Gap (eV)** from **Molecular structure based on SMILES (text and .xlsx)**. In the implementation, a machine learning algorithm called Unified Message Passing Model was trained for the prediction. It achieved **R² score of** for randomly selected 20% test dataset. In the following sections, you can check the details for the prediction results of this application, such as mean absolute error (MAE) and error distribution. Also, you can predict your own data using the trained model [here](#).

Contact E-mail : jang@kriict.kr
Status : Private
Created at : Jun 27, 2025
Edit Delete

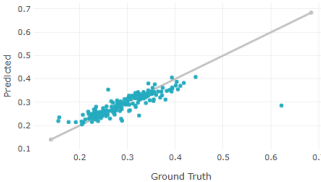
Trained Model and Prediction Results

Prediction results of the trained model on the randomly selected test dataset.

Prediction Model	MAE	RMSE	R ² Score	Actions
 Unified Message Passing Model Transformer-based graph neural network	0.0193	0.0338	0.6787	Download model Model architecture

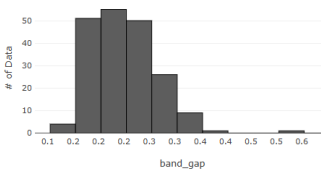
Prediction results

Prediction results of the trained model.



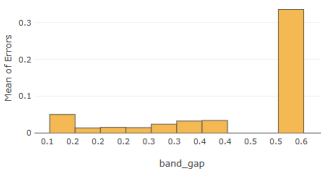
Target distribution

The number of data for each target range.



Error distribution

Mean of errors for each target range.



Prediction Using Model

Predict your input using the trained model. Your input should be uploaded as an Excel, XYZ, and OIF files.

Enter input values directly, or drag & drop one file here.

No	smiles	Gap (Prediction)
1	CCCCF	0.380

Add Row

Predict

Pre-trained 모델 활용 Prediction

ChemAI Use Case #1

Reaction condition optimization for non-oxidative conversion of methane using artificial intelligence *React. Chem. Eng.*, 2021, 6, 235–243

Input variable

- (1) Pressure (p)
- (2) Temperature (T)
- (3) Flow rate ($\dot{V}$)
- (4) H₂ content (H₂)
- (5) Length (l_R)
- (6) Diameter (d_R)

ML algorithm
(1) XGB
(2) FCNN

Target

- (1) C₂ yield (%)
- (2) C₂ selectivity (%)
- (3) Conversion (%)
- (4) Coke selectivity (%)

Hyperparameters
Number of nodes: 10, 20, 30, 40, 50, 60, 70
Number of hidden layers: 1, 2, 3, 4, 5
Optimizer: Adam, Adadelta, RMSProp
Activation function: ReLU, Sigmoid, Tanh

Hyper parameter optimization

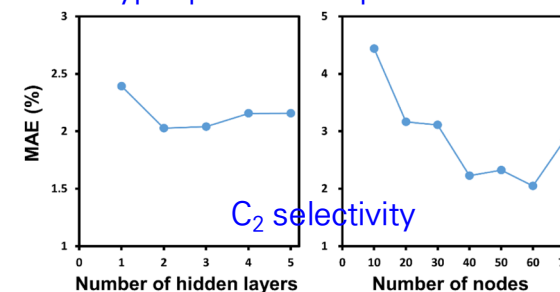
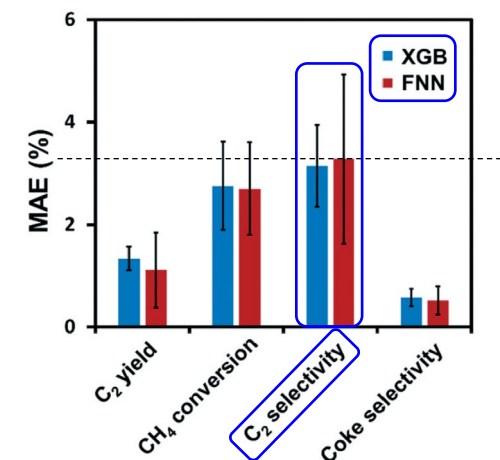


Table S1. Experimental data of MTOAH used in this study.

Index	Pressure (barg)	Temperature (°C)	Flow rate (sccm)	H ₂ content	Reactor length (cm)	Reactor diameter (mm)	CH ₄ conversion (%)	C ₂ selectivity (%)	Coke selectivity (%)	C ₂ yield (%)
1	0.031	890.1	40	0	45	16	2.07	41.01	12.79	0.85
2	0.069	920	80	0	45	16	1.89	46.81	16.06	0.89
3	0.024	890	120	0	45	16	2.55	37.84	35.13	0.97
4	0.114	939.9	120	0	45	16	2.56	44.39	26.57	1.14
5	0.032	900.2	40	0	45	16	3.57	38.28	15.61	1.37
6	0.05	919.9	60	0	45	16	3.69	38.93	31.05	1.44
7	0.07	930.1	80	0	45	16	3.96	39.48	25.16	1.56
8	0.024	899.9	120	0	45	16	4.42	39.08	28.01	1.73
9	0.116	950	120	0	45	16	5.16	39.48	25.36	2.04
10	0.032	910.1	40	0	45	16	6.02	35.36	21.56	2.13
11	0.367	1120	320	1	15	6.8	2.72	78.36	10.52	2.13
12	0.05	929.8	60	0	45	16	6.57	35.13	30.79	2.31

Numeric vector dataset, # of data : 254

Range of C₂ selectivity :
8.7 ~ 98.6 (%)



ChemAI Use Case #1

Step 1: Training Dataset Upload

Select input types of your training dataset. The data handling process for each input type is presented in [Documentation](#).

☒ Numeric Vector
 ☐ Chemical Formula
 ☐ SMILES
 ☐ Crystal Structure
 ☐ Image
 ☐ Spectrum

Example Dataset ▾ data_nocm.xlsx Browse

#	Feature Name	Category	Mean	Range	Correlation	Input	Target
1	pressure	Numerical	-0.546±0.000	[-0.004, 0.776]	-0.206	☒	☐
2	max_T	Numerical	0.405±0.000	[-0.500, 0.750]	0.799	☒	☐
3	flow_rate	Numerical	-0.201±0.000	[-0.118, 0.824]	0.214	☒	☐
4	H2(CH4+Ar)	Numerical	-0.288±0.000	[-0.273, 0.818]	0.238	☒	☐
5	reactor_len	Numerical	-0.481±0.000	[-0.750, 0.750]	0.500	☒	☐
6	reactor_dia	Numerical	0.019±0.000	[-0.493, 0.733]	0.506	☒	☐
7	Conversion(CH4)	Numerical	13.978±0.000	[1.891, 9.889]	0.178	☐	☐
8	X_C2H4	Numerical	42.315±0.000	[12.866, 8.559]	0.561	☐	☐
9	X_C2	Numerical	70.145±0.000	[10.007, 98.567]	1.000	☐	☒
10	X_Coke	Numerical	16.247±0.000	[0.096, 9.957]	0.088	☐	☐
11	Y_C2	Numerical	8.678±0.000	[0.851, 9.951]	-0.073	☐	☐
12	Y_C2H4	Numerical	5.183±0.000	[0.603, 9.781]	0.723	☐	☐

Imputation method Fill to mean Fill to zero Fill statistically

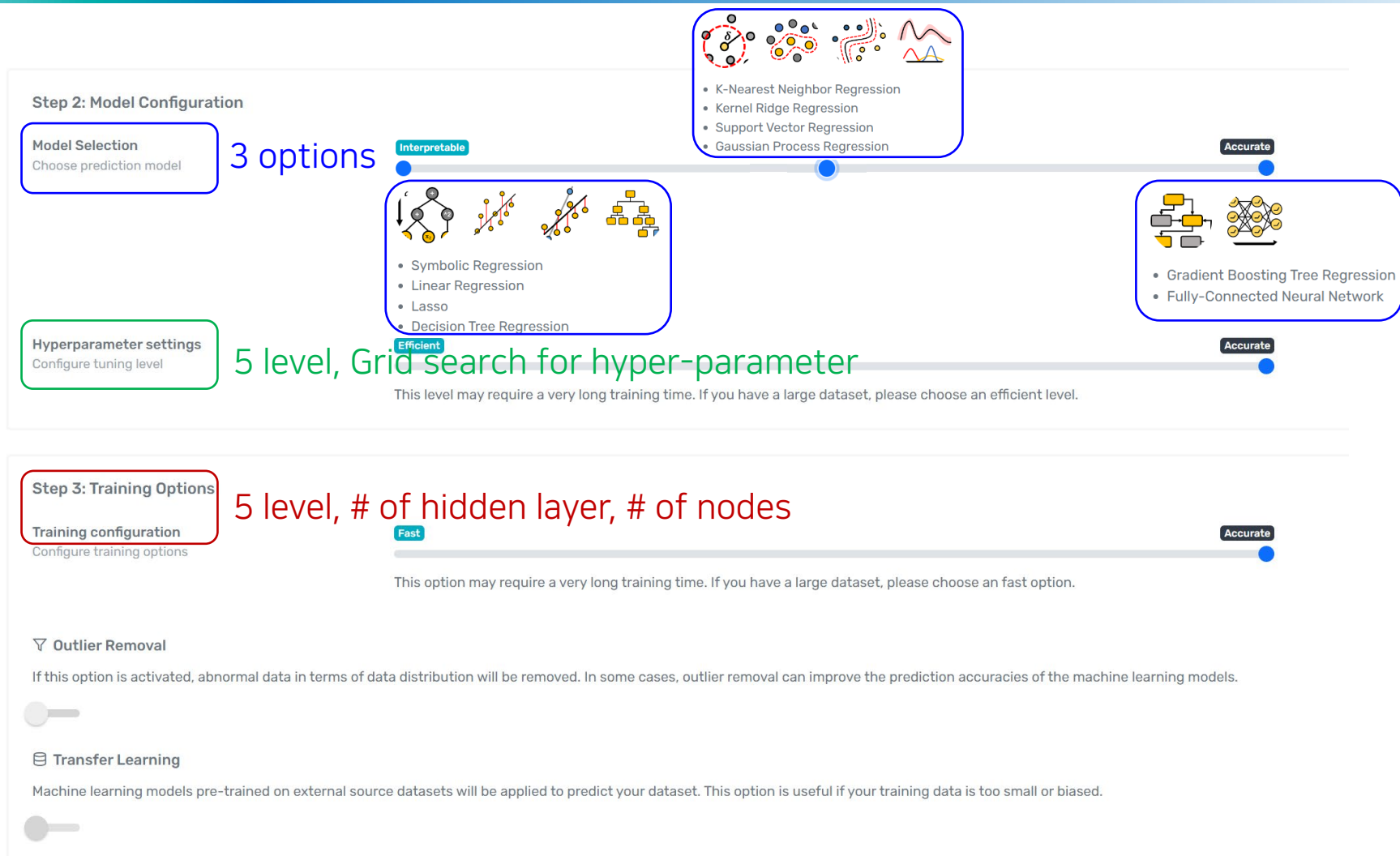
Replace missing inputs

6 inputs

• C₂ Selectivity

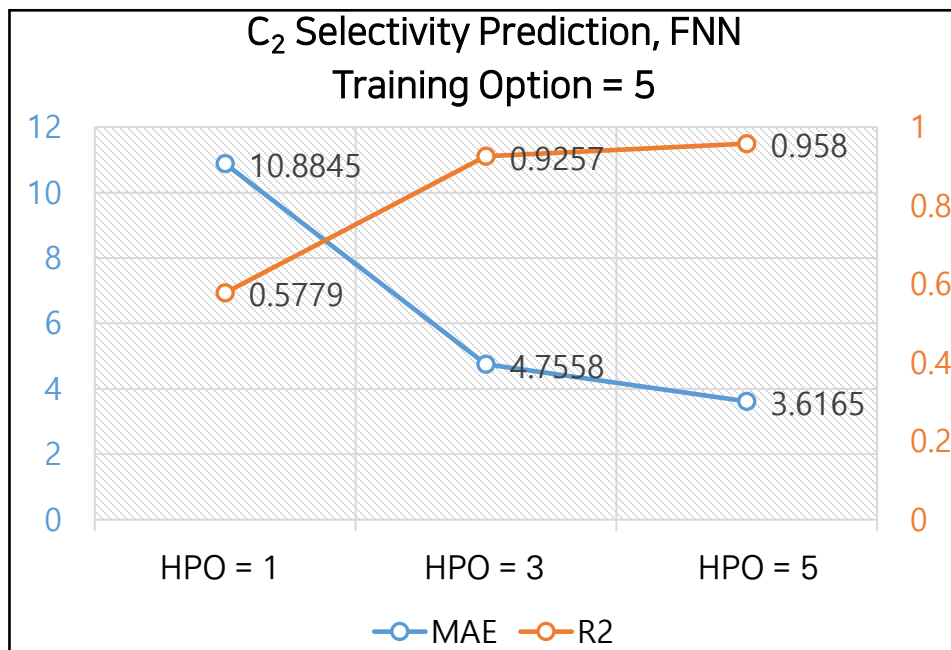
- 사용자 데이터셋 불러오기
- Feature(input) / Target 선택

ChemAI Use Case #1

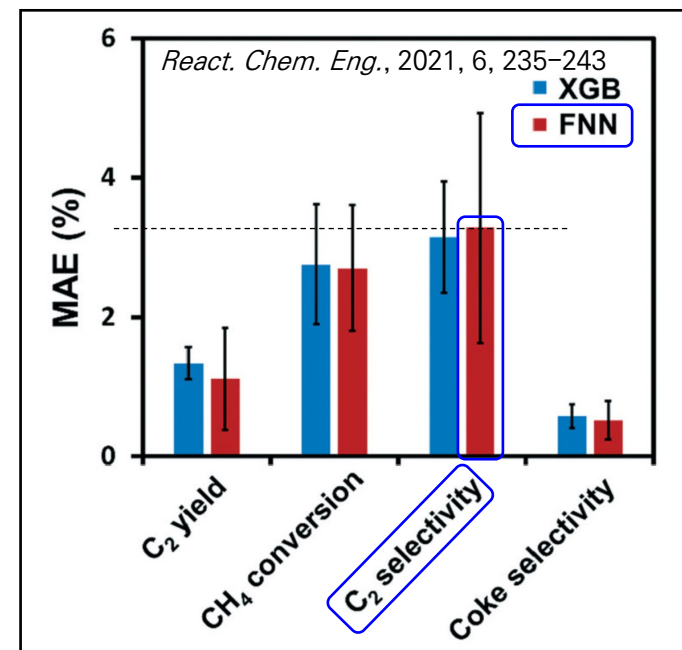


ChemAI Use Case #1

〈ChemAI 결과〉



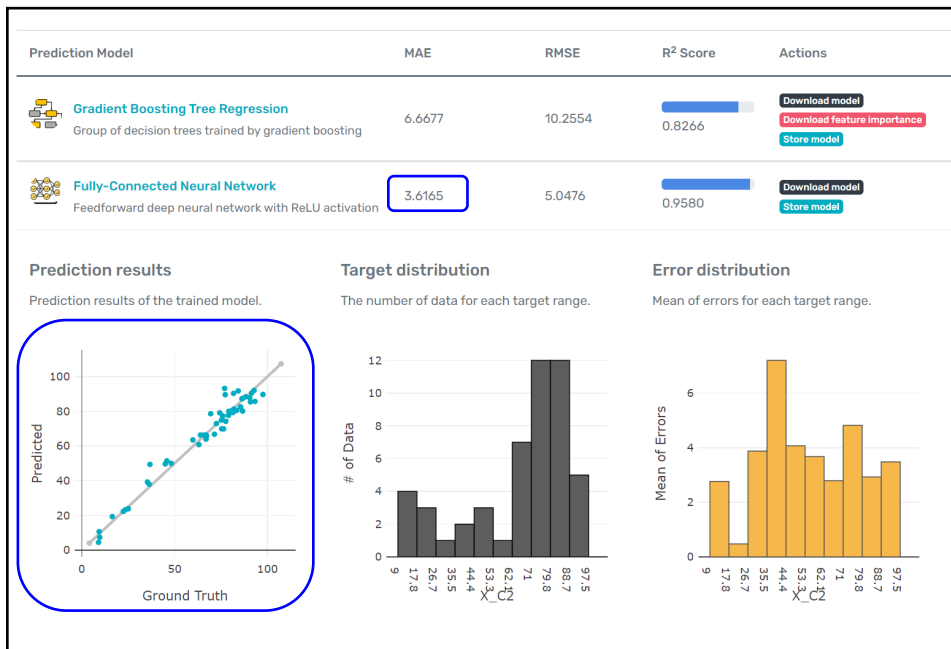
〈기존 논문 결과〉



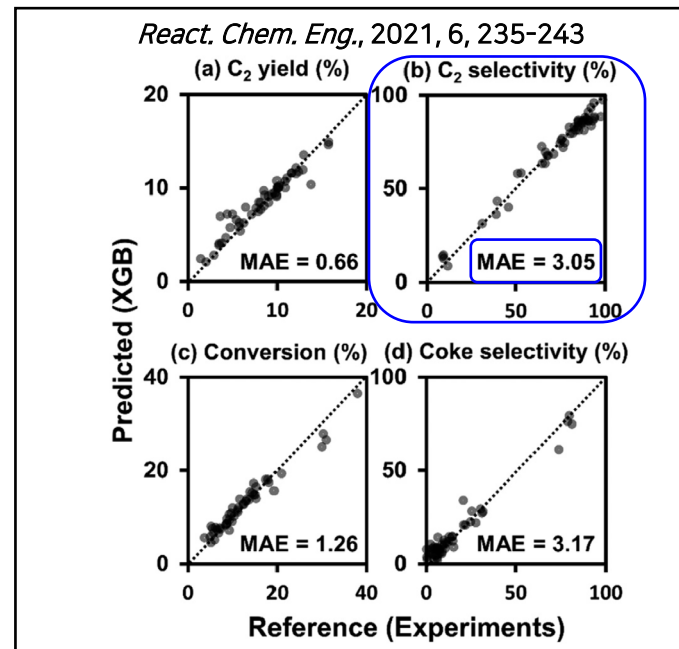
- Hyper-parameter 최적화 기능
- 매우 빠르게 최적 성능을 갖는 모델 및 Hyper-parameter 반환.
- 기존 논문에 보고된 결과와 비슷한 수준의 C₂ Selectivity (FNN) 예측 성능을 나타냄.

ChemAI Use Case #1

<ChemAI 결과>



<기존 논문 결과>



- 모델별 예측 성능 (MAE, RMSE, R² score)
- Scatter plot, Target distribution, Error distribution 정보 제공.
- ChemAI 결과 (FNN, MAE:3.616), 기본논문결과 (XGB, MAE:3.05)

ChemAI Use Case #2

Predicting the Band Gaps of Inorganic Solids by Machine Learning
J. Phys. Chem. Lett. 2018, 9, 7, 1668–1673

Input variable

Chemical Formula
 (a descriptor set based
 only on the
 elemental properties)

ML algorithm
 (1) SVR

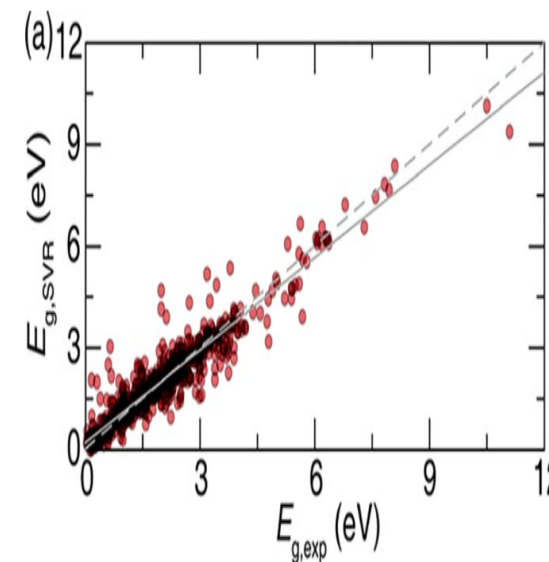
Target

Band gap (eV)

Chemical formula dataset
 # of data : 3895

composition	E_g (eV)
Hg _{0.7} Cd _{0.3} Te	0.35
CuBr	3.08
LuP	1.30
Cu ₃ SbSe ₄	0.40
ZnO	3.44
PtSb ₂	0.08
ZnIn ₂ S ₄	2.68
K ₂ Cd ₃ Te ₄	2.26
K ₄ Sn ₃ Ce ₃ S ₁₄	2.46
Pb ₆ B ₂ CrO ₁₂	1.80
InSb	0.22
Sr ₂ Be ₂ B ₂ O ₇	8.00
K ₂ CuNbS ₄	2.82
PrTe ₂	1.02
BAs	1.46
Ba ₄ Ga ₄ GeSe ₁₂	2.18
CuBr	2.94
HgSe	0.07
AgIn ₅ Te ₈	0.47
Rb ₇ Th ₂ P ₆ Se ₂₁	2.07
TiS	1.37
CdS	2.50

regressor	RMSE (eV)	r^2
SVR _{rbf}	0.45	0.90
SVR _{lin}	0.73	0.72
SVR _{pol}	0.46	0.89
KNN	0.54	0.85
KRR	0.72	0.74



ChemAI Use Case #2

Chemical Formula based atomic variable 자동생성
→ Feature set 활용

Step 1: Training Dataset Upload

Select input types of your training dataset. The data handling process for each input type is presented in [Documentation](#).

☐ Numeric Vector ☒ Chemical Formula ☐ SMILES ☐ Crystal Structure ☐ Image ☐ Spectrum

Example Dataset: jz8b00124_si_002.xlsx Browse

#	Feature Name	Category	Chemical Formula	Mean	Range	Correlation	Input	Target
1	composition	Categorical	☒					
2	Eg (eV)	Numerical		2.042±0.000	[0.020, 9.920]	1.000	☐	☒

Imputation method
Replace missing inputs

☒ Fill to mean ☐ Fill to zero ☐ Fill statistically

Step 2: Model Configuration

Model Selection
Choose prediction model

Interpretable Accurate



- K-Nearest Neighbor Regression
- Kernel Ridge Regression
- Support Vector Regression
- Gaussian Process Regression

Hyperparameter settings
Configure tuning level

Efficient Accurate

Step 3: Training Options

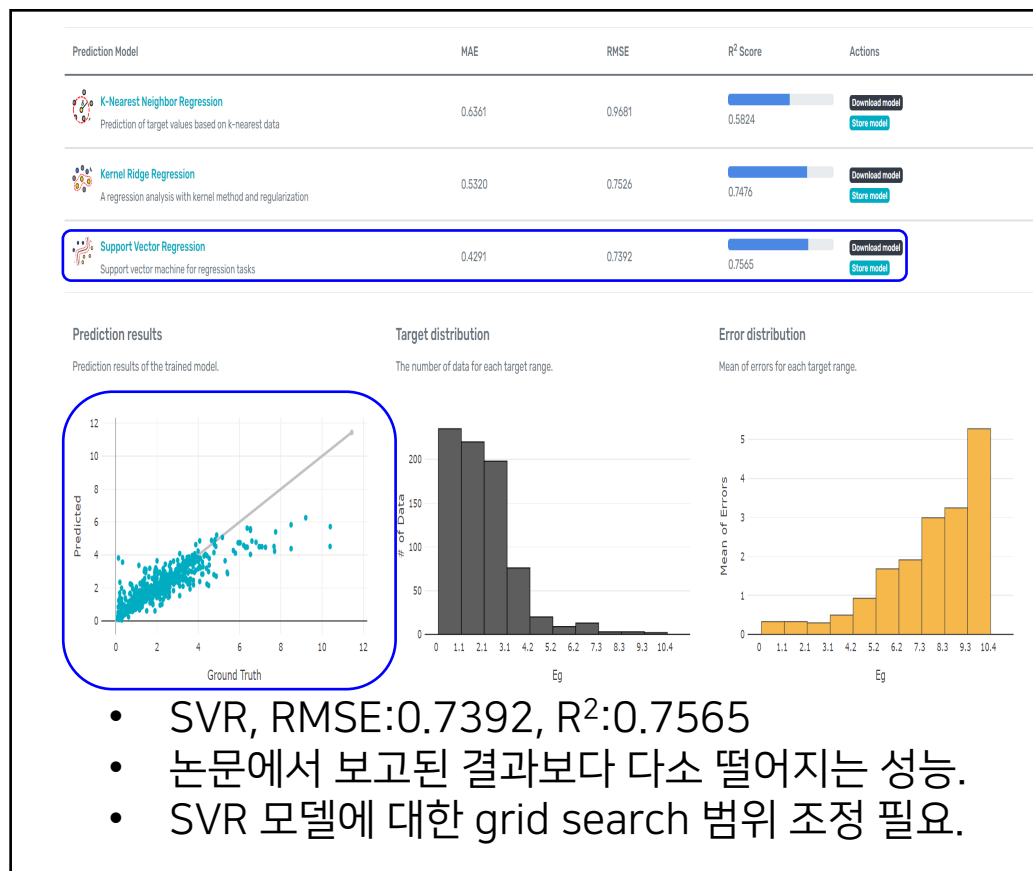
Training configuration
Configure training options

Fast Accurate

Band Gap 예측

ChemAI Use Case #2

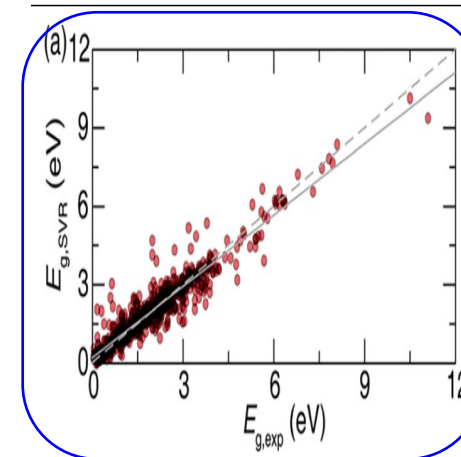
<ChemAI 결과>



<기존 논문 결과>

Predicting the Band Gaps of Inorganic Solids by Machine Learning *J. Phys. Chem. Lett.* 2018, 9, 7, 1668–1673

regressor	RMSE (eV)	r ²
SVR _{rbf}	0.45	0.90
SVR _{lin}	0.73	0.72
SVR _{pol}	0.46	0.89
KNN	0.54	0.85
KRR	0.72	0.74



Scientific Data 2020, 7, 295

Deep Learning Optical Spectroscopy Based on Experimental Database:
Potential Applications to Molecular Design JACS Au **2021**, 1, 427–438

Input variable

SMILES for Solvent
SMILSE for
Chromophore

ML algorithm
(1) GCN + FNN

Target

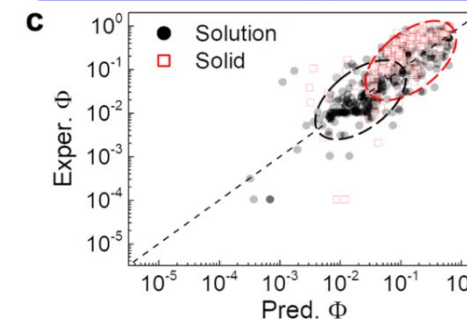
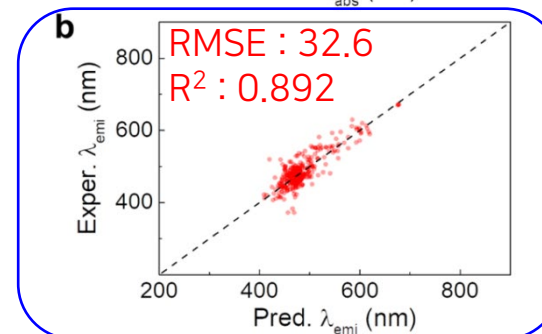
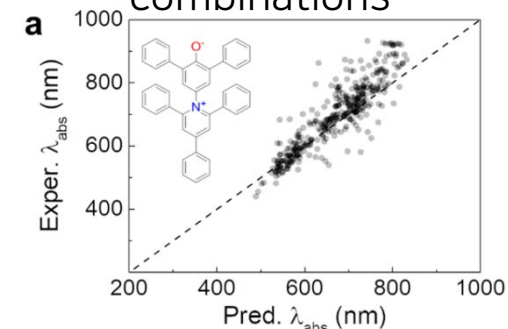
- absorption peak position
- emission peak position
- Bandwidth of absorption peak
- Bandwidth of emission peak
- quantum yield
- lifetime

Simplified molecular input
line entry system (SMILES)

	Chromophore	Solvent	Absorption max (nm)	Emission max (nm)	Lifetime (ns)	Quantum yield	log(ϵ /mol ⁻¹ dm ³ cm ⁻¹)	abs PWHM (cm ⁻¹)	emi PWHM (cm ⁻¹)	abs FWHM (nm)	emi FWHM (nm)
1	NHc1c1ccccc1Odc2cc1=O	O	355	410	2.804262479						
2	NHc1c1ccccc1Odc1cc2cc1=O	O	408	450	3.961965135						43
3	C1=O1cc2ccc3cc4cc5c6cc1ccccc3cc4cc5cc2cc1=O	ClC(Cl)Cl	526	535	3.602954423						
4	[O-]1c1c-c2n3ccc4cc2cc2ccc3ccc4cc1c2c34	CC#N	514	553.72	3.81						67.4
5	[O-]1c1c-c2n3ccc4cc2cc2ccc3ccc4cc1c2c34	C#C(O)O	524	555	4.7						50
6	CCN1CC1c1cc2cc1=Odc2c1=Odc2c1=O	C1CCCCC1	350	395	2.8					58	
7	CCN1CC1c1cc2cc1=Odc1=Odc2c1=O	CCOC(Cl)=O	361	416	3.1						
8	CCN1CC1c1cc2cc1=Odc1=Odc2c1=O	CC#N	367	430	3.4						
9	CCN1CC1c1cc2cc1=Odc1=Odc2c1=O	CCO	373	451	3.1						
10	CCN1CC1c1cc2cc1=Odc1=Odc2c1=O	OCC(O)CO	384	463	3.8						
11	CCN1CC1c1cc2cc1=Odc1=Odc2c1=Odc2c1=O	C1CCCCC1	376	413	4.1						
12	CCN1CC1c1cc2cc1=Odc1=Odc2c1=Odc2c1=O	CCOC(Cl)=O	392	479	4.6						
13	CCN1CC1c1cc2cc1=Odc1=Odc2c1=Odc2c1=O	CC#N	396	501	0.6						
14	CCN1CC1c1cc2cc1=Odc1=Odc2c1=Odc2c1=O	CCO	400	509	0.85						
15	CCN1CC1c1cc2cc1=Odc1=Odc2c1=Odc2c1=O	OCC(O)CO	413	524	2.7						
16	Cc1cc=O1cc2cc34cc1cc12CCCN4CCc3	C1CCCCC1	361	407	2.6						
17	Cc1cc=O1cc2cc34cc1cc12CCCN4CCc3	CC#N	380	450	3.3						
18	Cc1cc=O1cc2cc34cc1cc12CCCN4CCc3	CCO	387	473	4.5						
19	Cc1cc=O1cc2cc34cc1cc12CCCN4CCc3	O	396	489	5.9						
20	O=c1cc(C(F)(F)F)2cc34cc1cc12CCCN4CCc3	C1CCCCC1	393	455	4.3						
21	O=c1cc(C(F)(F)F)2cc34cc1cc12CCCN4CCc3	CCOC(Cl)=O	409	501	5.4						
22	O=c1cc(C(F)(F)F)2cc34cc1cc12CCCN4CCc3	CC#N	418	521	5.6						
23	O=c1cc(C(F)(F)F)2cc34cc1cc12CCCN4CCc3	CCO	421	531	3.4						
24	O=c1cc(C(F)(F)F)2cc34cc1cc12CCCN4CCc3	OCC(O)CO	434	546	3.5						
25	CN1CCc2cc3cc(C(F)(F)F)Fcc1=Odc3cc2c1	C1CCCCC1	382	439	4.2						
26	CN1CCc2cc3cc(C(F)(F)F)Fcc1=Odc3cc2c1	CC#N	405	510	5.6						
27	CN1CCc2cc3cc(C(F)(F)F)Fcc1=Odc3cc2c1	CCO	408	515	5.1						
28	Nc1ccccc1cc(C(F)(F)F)Fcc1=Odc1=O	CC#N	365	465	5.2						
29	CCN1CC1c1cc2cc1=Odc1=Odc2cc34cc1cc12CCCN4CCc3	CC#N	454	501	3.2						
30	CCN1CC1c1cc2cc1=Odc1=Odc2cc34cc1cc12CCCN4CCc3	CCO	403	480	2.8						
31	CCN1c1cc2cc1=Odc1=Odc2c1=O	CC#N	354	420	3.5						
32	CCN1c1cc2cc1=Odc1=Odc2c1=Odc2c1=O	CC#N	383	481	7.7						
33	3c1cc2cc3cc1=Odc1=Odc2cc34cc1cc12CCCN4CCc3	CO	530	555	3.7						
34	3c1cc2cc3cc1=Odc1=Odc2cc34cc1cc12CCCN4CCc3	O			1.52						
35	3c1cc2cc3cc1=Odc1=Odc2cc34cc1cc12CCCN4CCc3	O			1.63	0.95					

```
smiles dataset
# of data : 20235
```

30,094
chromophore/solvent
combinations



ChemAI Use Case #3

#	Feature Name	Category	SMILES	Mean	Range	Correlation	Input	Target
1	Tag	Numerical		9914.224±0.000	[1.000, 9999.000]	-0.341	☐	☐
2	Chromophore	Categorical	☒					
3	Solvent	Categorical	☒					
4	Emission max (nm)	Numerical		500.244±0.000	[1050.000, 924.000]	1.000	☐	☒
5	Molecular weight (g mol ⁻¹)	Numerical		476.865±0.000	[1000.118, 999.223]	0.485	☐	☐

Imputation method
Replace missing inputs

☒ Fill to mean
 ☐ Fill to zero
 ☐ Fill statistically

Emission peak position 선택

SMILES input → graph 형식변환
Node attribute, edge attribute, attention..

Step 2: Model Configuration

Model Selection
Choose prediction model

- Graph Convolutional Network
- Graph Attention Network
- Graph Isomorphism Network
- Graph Interaction Network of GCN

Hyperparameter settings
Configure tuning level

Efficient Accurate

Step 3: Training Options

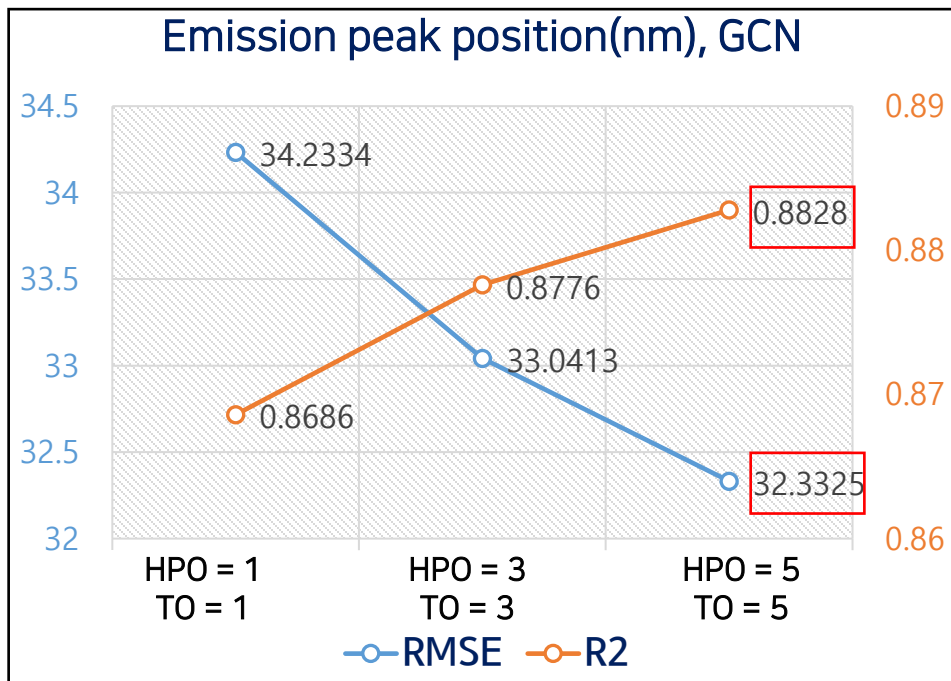
Training configuration
Configure training options

Fast Accurate

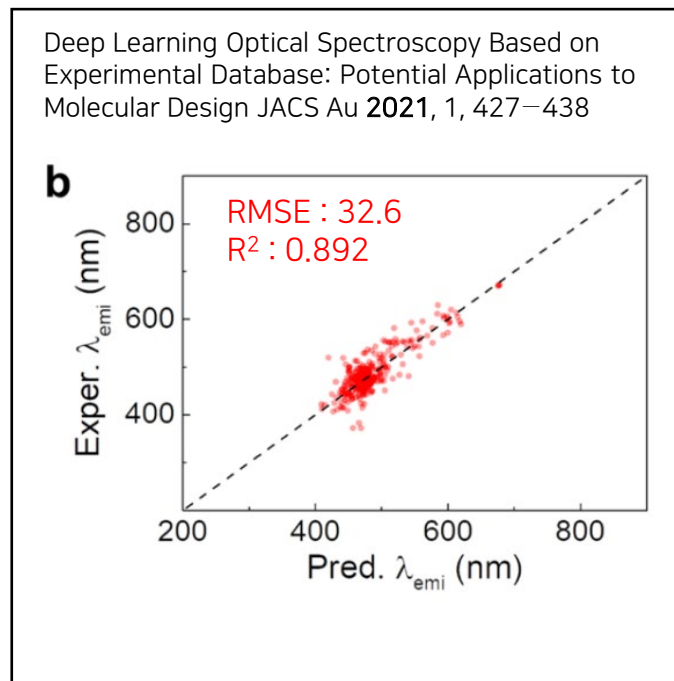
This option may require a very long training time. If you have a large dataset, please choose an fast option.

ChemAI Use Case #3

<ChemAI 결과>



<기존 논문 결과>



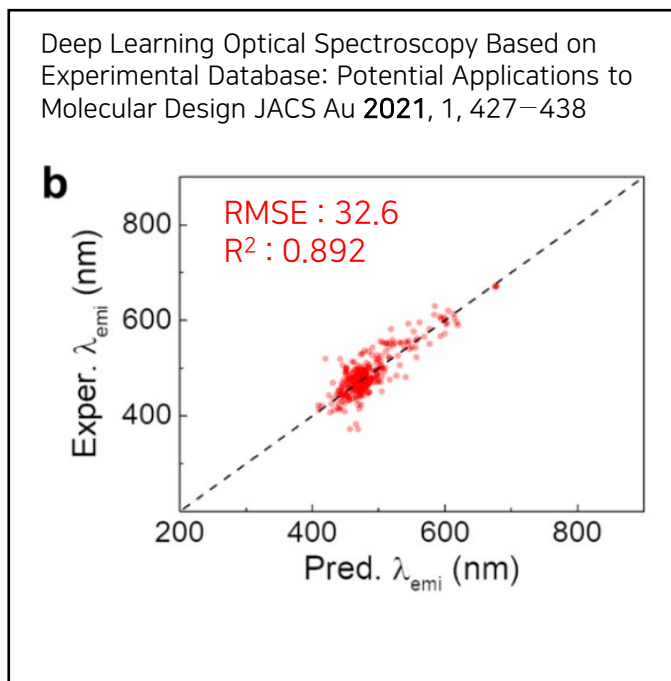
- 미세한 변화이지만, Hyper-parameter 최적화 기능 활용하여 최적 성능 탐색.
- 기존 논문에 보고된 결과보다 우수한 Emission peak position (GCN) 예측 성능을 나타냄.
- 20,000 개 정도의 Dataset에 대한 모델훈련도 ChemAI 플랫폼을 활용하여 가능. (P100, A100 지원)

ChemAI Use Case #3

<ChemAI 결과>



<기존 논문 결과>

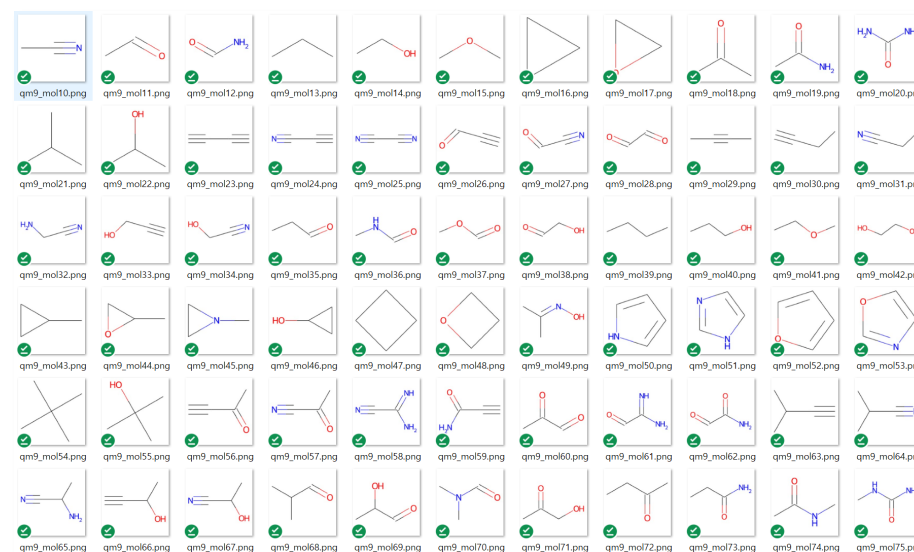


- ChemAI 결과 (GCN, RMSE:32.333), 기존 논문결과 (GCN, RMSE:32.6)
- 정량적으로 더 우수한 예측 성능.
- 분자정보를 활용한 고급기계학습 알고리즘(GCN)에 대한 지원 가능.

ChemAI Use Case #4

smiles	dipole_moment	isotropic_polarizability	homo	lumo	band_gap	electronic_spatial_extent	zpve	heat_capacity	internal_energy_OK	internal_energy_298K	enthalpy_298K	free_energy_298K
C	0	13.21	-0.3877	0.1171	0.5048	35.3641	0.044749	6.469	-40.47893	-40.476062	-40.475117	-40.498597
N	1.6256	9.46	-0.257	0.0829	0.3399	26.1563	0.034358	6.316	-56.525887	-56.523026	-56.522082	-56.544961
O	1.8511	6.31	-0.2928	0.0687	0.3615	19.0002	0.021375	6.002	-76.404702	-76.401867	-76.400922	-76.422349
C#C	0	16.28	-0.2845	0.0506	0.3351	59.5248	0.026841	8.574	-77.308427	-77.305527	-77.304583	-77.327429
C#N	2.8937	12.99	-0.3604	0.0191	0.3796	48.7476	0.016601	6.278	-93.411888	-93.40937	-93.408425	-93.431246
C=O	2.1089	14.18	-0.267	-0.0406	0.2263	59.9891	0.026603	6.413	-114.483613	-114.480746	-114.479802	-114.505268
CC	0	23.95	-0.3385	0.1041	0.4426	109.5031	0.074542	10.098	-79.764152	-79.760666	-79.759722	-79.787269
CO	1.5258	16.97	-0.2653	0.0784	0.3437	83.794	0.051208	8.751	-115.675816	-115.674872	-115.674872	-115.701876
CC#C	0.7156	28.78	-0.2609	0.0613	0.3222	177.1963	0.05541	12.482	-116.609549	-116.60555	-116.604606	-116.633775
CC#N	3.8266	24.45	-0.3264	0.0376	0.364	160.7223	0.045286	10.287	-132.71815	-132.714563	-132.713619	-132.742149
CC=O	2.5682	25.11	-0.254	-0.0198	0.2342	166.9728	0.055355	11.219	-153.787612	-153.783728	-153.782784	-153.812518
C(=O)N	3.7286	21.57	-0.2543	0.0302	0.2845	145.3078	0.045279	10.89	-169.860788	-169.856903	-169.855958	-169.885594
CCC	0.0597	34.75	-0.323	0.0949	0.4179	227.1361	0.103182	14.84	-119.052475	-119.047927	-119.046983	-119.078157
CCO	1.4131	27.87	-0.2619	0.0798	0.3417	193.1659	0.079754	13.546	-154.968412	-154.967467	-154.967467	-154.998148
COC	1.1502	28.13	-0.2525	0.091	0.3435	187.1015	0.079534	12.934	-154.960361	-154.956045	-154.9551	-154.985747
C1CC1	0.0005	30.82	-0.2888	0.1042	0.393	155.8145	0.081231	11.041	-117.824798	-117.821426	-117.820482	-117.849087
C1CO1	1.7675	24.04	-0.2682	0.1042	0.3724	129.891	0.057289	9.176	-153.742562	-153.73941	-153.738466	-153.766642
CC(=O)C	2.7362	35.53	-0.2431	-0.0087	0.2344	292.4367	0.083382	16.893	-193.082969	-193.082024	-193.082024	-193.116476
CC(=O)N	3.6367	31.83	-0.2436	0.0347	0.2783	267.6148	0.07319	16.561	-209.159302	-209.15402	-209.153076	-209.187468
C(=O)N(N)	3.4869	28.07	-0.2495	0.0556	0.3051	244.2308	0.063824	15.292	-225.221461	-225.217075	-225.216131	-225.247724
CC(C)C	0.0897	45.46	-0.3167	0.0843	0.401	355.0621	0.131146	20.273	-158.342346	-158.336603	-158.335658	-158.370016
CC(C)O	1.4259	38.58	-0.2612	0.074	0.3351	318.3721	0.107673	19.052	-194.267232	-194.260804	-194.259804	-194.294663
C#CC#C	0	38.52	-0.2599	-0.0214	0.2386	278.6264	0.097354	15.312	-153.459846	-153.455442	-153.454498	-153.482621
C#CC#N	3.792	32.66	-0.3102	-0.0543	0.2559	260.1996	0.027259	12.93	-169.557758	-169.553764	-169.55282	-169.581024
C(N)C#N	0.0023	27.7	-0.3696	-0.0926	0.277	242.9308	0.015951	10.398	-185.648533	-185.644825	-185.64388	-185.676552
C#CC=O	2.7824	31.14	-0.2777	-0.0735	0.2042	268.3921	0.037208	13.049	-190.624631	-190.620363	-190.619419	-190.650543
C(=O)C#N	2.3112	26.25	-0.3166	-0.11	0.2066	251.0007	0.02654	11.329	-206.717858	-206.717875	-206.716931	-206.747625
C(=O)C=O	0.002	26.12	-0.2668	-0.1113	0.1555	266.8164	0.036943	12.147	-227.798785	-227.79457	-227.793626	-227.825074

QM9 dataset 활용.
 Python rdkit 활용.
 PNG type image 생성.
 파일명은 qm_mol###.png
 Smiles + image : 멀티모달러닝 테스트 진행 (예측 포함)
 ex) chemical formula + SEM image



코딩 못하는 사람도 OK, 국내 최초 화학분야

IT/과학

최신뉴스 | 과학 | 디지털 | 컴퓨터/인터넷 | 뉴미디어/통신 | 게임 | IT/과학 일반

코딩 못하는 사람도 OK...국내 최초 화학분야 'AI' 첫선

헤럴드경제 원문 | 기사전송 2021-08-30 12:02



| - 화학분야, 화학 분야 인공지능 플랫폼 'ChemAI' 서비스 출시



인공지능 플랫폼 'ChemAI' 서비스를 개발한 연구자들. 왼쪽부터 장승훈 박사, 장현주 센터장, 나경석 연구원.[한국화학연구원 제공]

[헤럴드경제=구본혁 기자] 한국화학연구원 화학데이터기반연구센터는 사용자의 데이터를 자유롭게 활용해 인공지능(AI)을 연구에 활용할 수 있는 웹기반 인공지능 플랫폼 'ChemAI'를 구축했다고 30일 밝혔다.

IT·과학

코딩 못하는 사람도 OK...국내 최초 화학분야 'AI' 첫선

입력 2021-08-30 12:01:32

- 화학분야, 화학 분야 인공지능 플랫폼 'ChemAI' 서비스 출시
- 알고리즘 자동추천 기능, 하이퍼파라미터 최적화 자동 수행



인공지능 플랫폼 'ChemAI' 서비스를 개발한 연구자들. 왼쪽부터 장승훈 박사, 장현주 센터장, 나경석 연구원.[한국화학연구원 제공]

[헤럴드경제=구본혁 기자] 한국화학연구원 화학데이터기반연구센터는 사용자의 데이터를 자유롭게 활용해 인공지능(AI)을 연구에 활용할 수 있는 웹기반 인공지능 플랫폼 'ChemAI'를 구축했다고 30일 밝혔다.



무기형광체 데이터셋 수집 및 광학특성 예측

Optical property dataset of inorganic phosphor
Sci. Rep. (2024) 14 7639

**Maximum PL emission wavelength Prediction**

- Composition
- Temperature
- Wavelength of excitaion source

Maximum PL emission wavelength (nm)

0.843

Trained modelSource dataset

CompositionBa2Bi0.95V3O11Eu0.05

Temperature298

Wavelength of excitaion source394

Predict

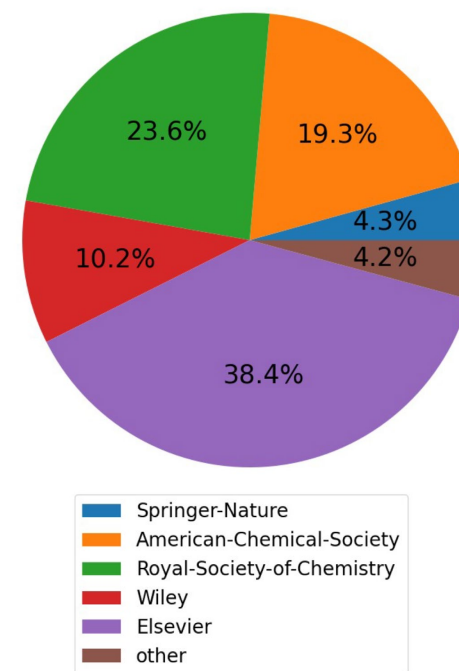
☞ Predicted value: 594.34839

ChemAI – Maximum PL emission wavelength, PL decay time,
CIE coordinate prediction Predictions

Data collection

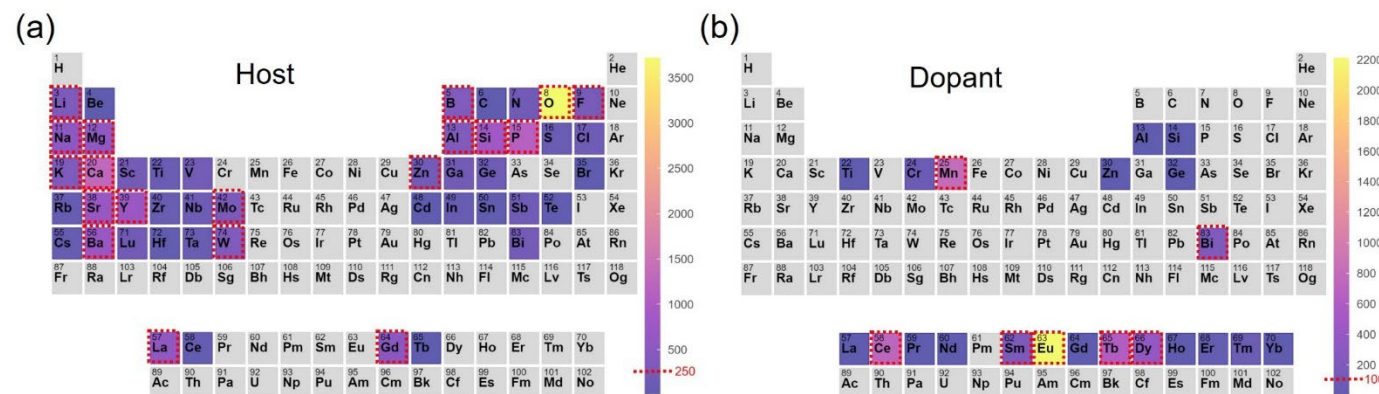
Construction of an inorganic phosphorus dataset.

- From [553 articles](#) on inorganic phosphors directly downloaded from various verified publishers.
- Searched for articles using keywords such as "PL", "PL excitation", "decay time", "quantum efficiency", and "thermal quenching temperature"
- Composition information and optical property information on a total of [3,952 inorganic phosphors](#)
- limited to cases involving up to two dopants



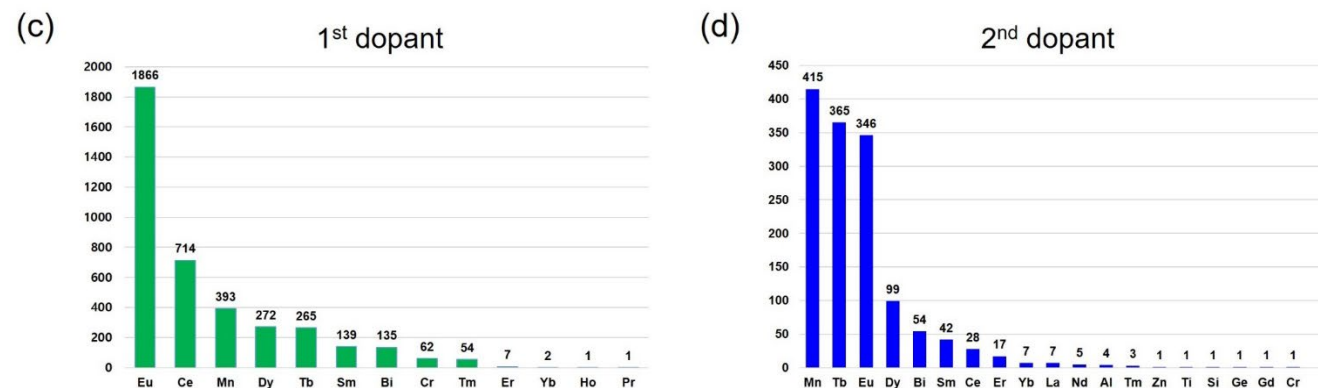
Data collection

Element frequencies for the (a) **hosts** and (b) **dopants**



- **Host:** oxide, phosphide, boride, silicate, and fluoride
- Alkali metals such as Li, Na, K, Mg, Ca
- Transition metals such as Y, Mo, W, and Zn
- **Dopant:** rare earth elements such as Eu, Sm, Tb, and Dy and elements such as Mn and Bi
- 21 dopant elements

Histograms of the elements comprising the (c) first and (d) second dopants



- Eu appears most frequently.
- REEs such as Ce, Sm, Tb, and Dy are also utilized.

Data Records

Inorganic phosphor optical property dataset: IPOP dataset

Host info.			Dopant (up to 2 allowed) info.			Target properties (w/ measurement condition) info.										Publication info.						
Tag	Inorganic phosphor	Host	1 st dopant	1 st dopant valency	1 st doping concentration	2 nd dopant	2 nd dopant valency	2 nd doping concentration	Emission max. (nm)	Excitation max. (nm)	Temp. (K)	Log ₁₀ Decay time (ms)	Decay time (ms)	CE's coordinate	CE's coordinate	Int. quantum efficiency (%)	Ext. quantum efficiency (%)	Thermal quenching temp. (K)	Excitation source (nm)	PLC detection energy (meV)	Reference	Publisher
1	Ba2SiO4:Pr3+/D01	Ba2SiO4	Pr3+	3	0.05				619	384	298	6.017013339	100000						384	619	https://doi.org/10.1038/nphoton.2010.04.012	EBV
2	Ba2SiO4:Y3+/D01	Ba2SiO4	Y3+	3	0.2				619	384	298	6.017013339	100000						384	619	https://doi.org/10.1038/nphoton.2010.04.012	EBV
3	Ba2SiO4:Y3+/D01	Ba2SiO4	Y3+	3	0.4				619	384	298	6.017013339	100000						384	619	https://doi.org/10.1038/nphoton.2010.04.012	EBV
4	Ba2SiO4:Y3+/D01	Ba2SiO4	Y3+	3	0.8				619	384	298	5.96211235	90000					451	384	619	https://doi.org/10.1038/nphoton.2010.04.012	EBV
5	Ba2SiO4:Y3+/D01	Ba2SiO4	Y3+	3	0.8				619	384	298	5.95814865	70000						384	619	https://doi.org/10.1038/nphoton.2010.04.012	EBV
6	Ba2SiO4:Y3+/D01	Ba2SiO4	Y3+	3	1				619	384	298	5.97213601	100000						384	619	https://doi.org/10.1038/nphoton.2010.04.012	EBV
7	CuAl4Si8O20:Pr3+/D01	CuAl4Si8O20	Pr3+	3	0.05				613	298	298			0.544	0.455	13.2			384	613	https://doi.org/10.1038/nphoton.2010.04.012	EBV
8	CuAl4Si8O20:Pr3+/D01	CuAl4Si8O20	Pr3+	3	0.08				613	298	298			0.564	0.457	16.1			384	613	https://doi.org/10.1038/nphoton.2010.04.012	EBV
9	CuAl4Si8O20:Pr3+/D01	CuAl4Si8O20	Pr3+	3	0.15				613	298	298			0.579	0.483	18.8			384	613	https://doi.org/10.1038/nphoton.2010.04.012	EBV
10	CuAl4Si8O20:Pr3+/D01	CuAl4Si8O20	Pr3+	3	0.35				613	395	298			0.615	0.388	15.8			384	613	https://doi.org/10.1038/nphoton.2010.04.012	EBV
11	CuAl4Si8O20:Pr3+/D01	CuAl4Si8O20	Pr3+	3	0.65				613	298	298			0.607	0.393	12.6			384	613	https://doi.org/10.1038/nphoton.2010.04.012	EBV
12	CuAl4Si8O20:Pr3+/D01	CuAl4Si8O20	Pr3+	3	0.55				613	298	298			0.594	0.405	10.1			384	613	https://doi.org/10.1038/nphoton.2010.04.012	EBV
13	Si6Mg6P14O40:Pr3+/D01	Si6Mg6P14O40	Pr3+	3	0.01				548	378	298	0.08105805	1.17	0.205	0.186				378	548	https://doi.org/10.1038/nphoton.2010.04.012	RSC
14	Si6Mg6P14O40:Pr3+/D01	Si6Mg6P14O40	Pr3+	3	0.06				548	378	298	0.255	0.175						378	548	https://doi.org/10.1038/nphoton.2010.04.012	RSC
15	Si6Mg6P14O40:Pr3+/D01	Si6Mg6P14O40	Pr3+	3	0.1				548	378	298	0.01189209	1.05	0.238	0.303				378	548	https://doi.org/10.1038/nphoton.2010.04.012	RSC
16	Si6Mg6P14O40:Pr3+/D01	Si6Mg6P14O40	Pr3+	3	0.15				548	378	298	0.265	0.176						378	548	https://doi.org/10.1038/nphoton.2010.04.012	RSC
17	Si6Mg6P14O40:Pr3+/D01	Si6Mg6P14O40	Pr3+	3	0.14				548	378	298	0.02098611	1.73	0.245	0.441				378	548	https://doi.org/10.1038/nphoton.2010.04.012	RSC
18	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.01				453	420	298	2.61213857	430						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
19	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.01				453	420	312	2.6224929	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
20	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.01				453	420	312	2.61412078	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
21	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.01				453	420	413	2.63144649	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
22	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.01				453	420	413	2.6224929	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
23	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	420	311	2.59104407	399	0.201	0.16				378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
24	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	420	311	2.60320991							378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
25	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	420	312	2.63144649	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
26	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	420	312	2.63212124	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
27	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	420	413	2.6224929	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
28	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	420	413	2.6224929	420						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
29	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.07				453	380	298	2.47131031	280						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
30	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.07				453	380	312	2.48217088	280						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
31	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.07				453	380	312	2.47131035	280						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
32	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.07				453	380	422	2.5031394	320						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
33	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.07				453	380	422	2.50313997	320						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
34	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	0.91210100	830						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
35	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.54801361	15.35						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
36	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
37	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
38	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
39	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
40	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
41	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
42	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
43	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
44	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
45	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
46	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
47	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
48	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
49	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015	13.24						378	453	https://doi.org/10.1038/nphoton.2010.04.012	SN
50	Na2HfSi2P2O10:Pr3+/D01	Na2HfSi2P2O10	Pr3+	3	0.03				453	320	298	1.52108015</										

Data Records

No.	Column name	Unit	Data type	Description
1	Tag	—	Float	The numbering of data points
2	Inorganic phosphor	—	String	Composition of phosphor
3	Host	—	String	Host composition of phosphor
4	1 st dopant	—	String	1 st dopant atom of phosphor
5	1 st dopant valency	—	Float	Valency of 1 st dopant atom
6	1 st dopant concentration	—	Float	Doping concentration of 1 st dopant
7	2 nd dopant	—	String	2 nd dopant atom of phosphor
8	2 nd dopant valency	—	Float	Valency of 2 nd dopant atom
9	2 nd dopant concentration	—	Float	Doping concentration of 2 nd dopant
10	Temp. (K)	K	Float	Measurement temperature
11	Emission max. (nm)	nm	Float	Maximum PL emission wavelength, $\lambda_{PL,max}$
12	CIE X coordinate	—	Float	X coordinate in CIE chromaticity diagram, X_{CIE}
13	CIE Y coordinate	—	Float	Y coordinate in CIE chromaticity diagram, Y_{CIE}
14	Int. quantum efficiency (%)	—	Float	Internal quantum efficiency, QE_{int}
15	Ext. quantum efficiency (%)	—	Float	External quantum efficiency, QE_{ext}
16	Thermal quenching temp. (K)	K	Float	Thermal quenching temperature, T50
17	Excitation source (nm)	nm	Float	Wavelength of excitation source, λ_{exc}
18	1st excitation max. (nm)	nm	Float	1st Maximum PLE wavelength, $\lambda_{PLE,max(1st)}$
19	2nd excitation max. (nm)	nm	Float	2nd Maximum PLE wavelength, $\lambda_{PLE,max(2nd)}$
20	3rd excitation max. (nm)	nm	Float	3rd Maximum PLE wavelength, $\lambda_{PLE,max(3rd)}$
21	Log ₁₀ Decay time (log ₁₀ [ns])	log ₁₀ [ns]	Float	Logarithmic value of PL decay time, $\tau_{log,PL}$
22	Decay time (ns)	ns	Float	Photoluminescence decay time, τ_{PL}
23	Monitoring energy (nm)	nm	Float	Decay time and PLE monitoring wavelength, λ_{mon}
24	Reference	—	String	Source document DOI
25	Publisher	—	String	Abbreviation for journal publisher
26	MP-ID	—	String	Materials project ID

Table 1. Description of the inorganic phosphor dataset.

- (1) Material information of the host material and dopant element
- (2) Measurement conditions such as wavelength of excitation and monitoring energy
- (3) Target properties of inorganic phosphors including maximum emission position, decay time, and quantum efficiency
- (4) Meta-information such as source document digital object identifier (DOI), tag number, and materials project (MP) ID

Ruthenium
Example: $\text{Lu}_3\text{Al}_{4.994}\text{O}_{12}\text{Mn}_{0.003}\text{Mg}_{0.003}$

Tag	745
Inorganic phosphor	$\text{Lu}_3\text{Al}_{4.994}\text{O}_{12}\text{Mn}_{0.003}\text{Mg}_{0.003}$
Host	$\text{Lu}_3\text{Al}_{4.994}\text{O}_{12}$
1 st dopant	Mn
1 st dopant valency	4
1 st dopant concentration	0.003
2 nd dopant	Mg
2 nd dopant valency	2
2 nd dopant concentration	0.003
Temp. (K)	298
Emission max. (nm)	670
CIE x coordinate	0.713
CIE y coordinate	0.286
Int. quantum efficiency (%)	72.41
Ext. quantum efficiency (%)	41.39
Thermal quenching temp. (K)	NaN
Excitation source (nm)	326
Excitation max. (nm)	326
Log ₁₀ Decay time (log ₁₀ [ns])	6.094122
Decay time (ns)	1242000
Monitoring energy (nm)	670
Reference	https://doi.org/10.1039/C7TC02514A
Publisher	RSC
MP-ID	mp-14132

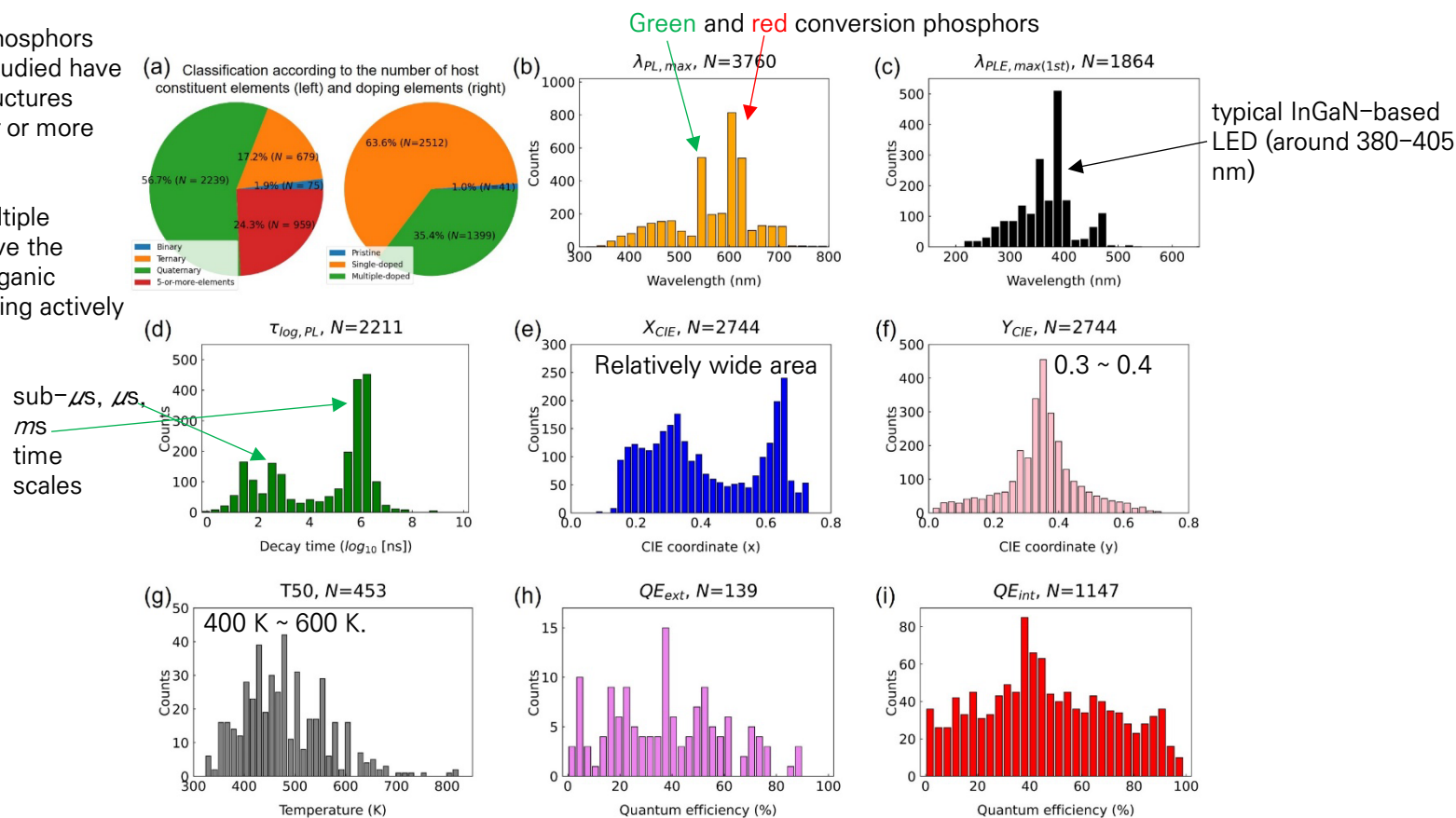
Table 2. Optical properties of $\text{Lu}_3\text{Al}_{4.994}\text{O}_{12}\text{Mn}_{0.003}\text{Mg}_{0.003}$.

Data Records

Distribution of the material and optical properties of the inorganic phosphors

Most inorganic phosphors currently being studied have complex host structures composed of four or more elements.

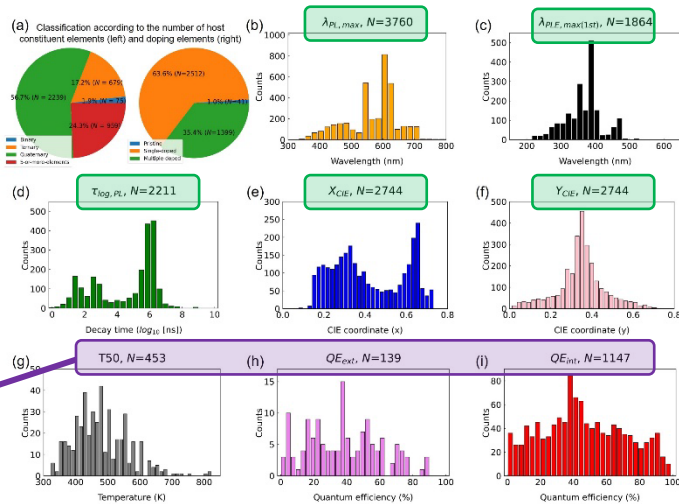
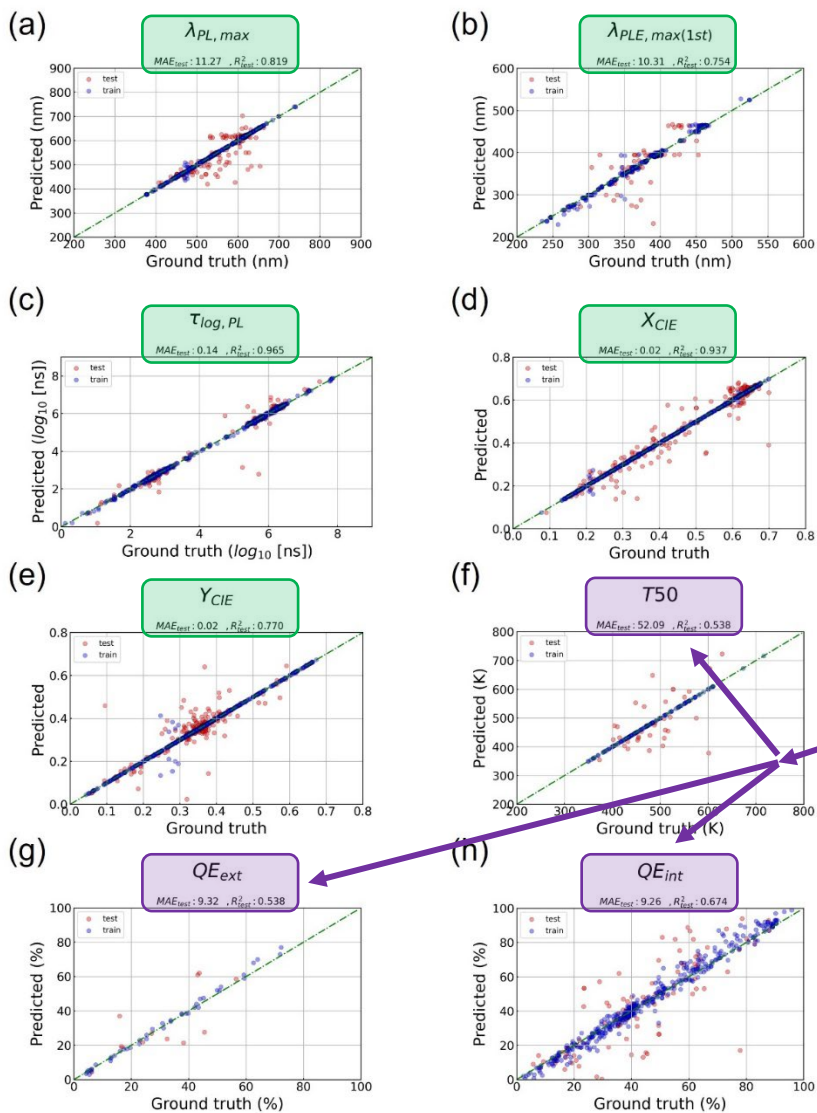
Studies using multiple dopants to improve the properties of inorganic phosphors are being actively conducted.



157 QE_{ext} and 1,168 QE_{int} data points reported in % units

Data-Driven Optical Property Prediction

The best prediction model for each target.



Summary

- For data-driven research on inorganic phosphors, we conducted a literature search to construct a materials dataset that contains experimentally synthesized inorganic phosphors and their experimentally measured physical properties.
- We collected eight essential physical properties of inorganic phosphors, such as photoluminescence maximum wavelengths, CIE coordinates, lifetime, and quantum efficiency. (16,023 observations of 2,238 host materials)
- We also collected the measurement conditions to ensure the reliability of the dataset.
- We conducted an ML experiment using the inorganic phosphor optical property dataset to obtain preliminary ML results.

Optical property dataset of inorganic phosphor
[Seunghun Jang*](#), GS Na, Y Choi, H Chang, Sci. Rep. 2024, 14, 7639



전이학습 활용 고분자 유리전이온도 예측

High Glass Transition Temperature Fluorinated Polymers Based on Transfer Learning with Small Experimental Data
Macromol. Rapid Commun. (2024) 45, 2400161

 **Glass transition temperature prediction**

• Monomer structure (SMILES)

Temperature (K)

Monomers

CC(=C)C(=O)OCC(F)(F)C(F)(F)C(F)(F)C(F)F 4
CCCCCCCCCCCCCCCCOC(=O)C(C)=C 4
CC(=C)C(=O)OC12CC3CC(C1)CC(C3)(C2)O 2

Predict

Predicted value: 36.20281 (K)

ChemAI – Glass transition temperature prediction

Fluorinated Polymer Data platform: F-polymer

<https://f-polymer.chemdx.org/>

Statistics Search Data Visualization DB toCSV Manage
Seunghun Jang

- Data 메뉴, 450여개 불소 고분자 data에 대한 개략적인 정보 표시
- 현재 일부 데이터 공개 (회원가입 필요)

+
Show filters
🔍

실험 이름

Exp No.	Id	Exp_name
0416	FA-00595	0416_HPFBA-CO-6-90
0413	FA-00591	0413_PFPFA-CO-6-90
0097	FA-00410	0097_PFPFA-CO-6
0419	FA-00598	0419_NFHA-CO-6-noacid-90
0104	FA-00208	0104_NFHA-CO-6-nooh
0103	FA-00207	0103_NFHA-CO-6-noacid
0132	FA-00246	0132_NFHA-60-10-0-30
0198	FA-00389	0198_NA31M312
0460	FA-00645	0460_HxFBA-HEMA82-90
0144	FA-00488	0144_HxFBA-HEMA82

모노머, Feed ratio

Monomer_a	Monomer_b	Monomer_c	Monomer_d	Monomer_e
HPFBA / 6	SMA / 1	MMA / 0	MAA / 1	HEMA / 2
PFPFA / 6	SMA / 1	MMA / 0	MAA / 1	HEMA / 2
PFPFA / 6	SMA / 1	MMA / 0	MAA / 1	HEMA / 2
NFHA / 6	SMA / 1	MMA / 1	MAA / 0	HEMA / 2
NFHA / 6	SMA / 1	MMA / 2	MAA / 1	HEMA / 0
NFHA / 6	SMA / 1	MMA / 1	MAA / 0	HEMA / 2
NFHA / 6	SMA / 1	MMA / 0	MAA / 0	HEMA / 3
NFHA / 3	SMA / 1	MA / 3	AA / 1	HEA / 2
HxFBA / 8	SMA /	MAA /	MAA /	HEMA / 2
HxFBA / 8	SMA /	MAA /	MAA /	HEMA / 2

고분자 물성

NMR	DSC Tg (degC) ↑	TGA 1% loss (degC)	Surface E (mN/m)	1 kHz Dielect. Const.
X	-29.27	191.01	14.94	1.885
X	-29.23	201.87	15.14	1.857
O	-29.23	201.87	15.14	2.64128
X	-29.22	212.62	X	2.501
O	-29.22	214.75	5.9	2.2005
O	-29.22	212.62	10.7	2.1355
O	-29.1	274.69	14.7	2.1962
O	-14.7	242.47	10.52	2.2115
X	-14.56	245.97	X	2.333
O	-14.56	245.97	X	2.0515

날짜

Created_at	Updated_at
2022-09-08 01:42:56	2022-09-26 05:46:45
2022-09-08 01:29:15	2022-09-26 05:44:57
2022-02-21 06:58:28	2022-09-22 07:16:49
2022-09-08 01:46:36	2022-09-26 05:47:48
2020-10-29 08:15:24	2022-09-22 07:16:49
2020-10-29 08:12:47	2022-09-22 07:16:48
2021-03-02 10:12:54	2022-09-22 07:16:49
2021-12-01 09:42:45	2022-06-03 05:58:22
2022-09-15 07:15:16	2022-09-15 07:17:16
2022-05-31 08:13:30	2022-07-01 08:14:25

사용자 편집 가능

✖

✖

✖

✖

✖

✖

✖

✖

✖

Exp No. 클릭

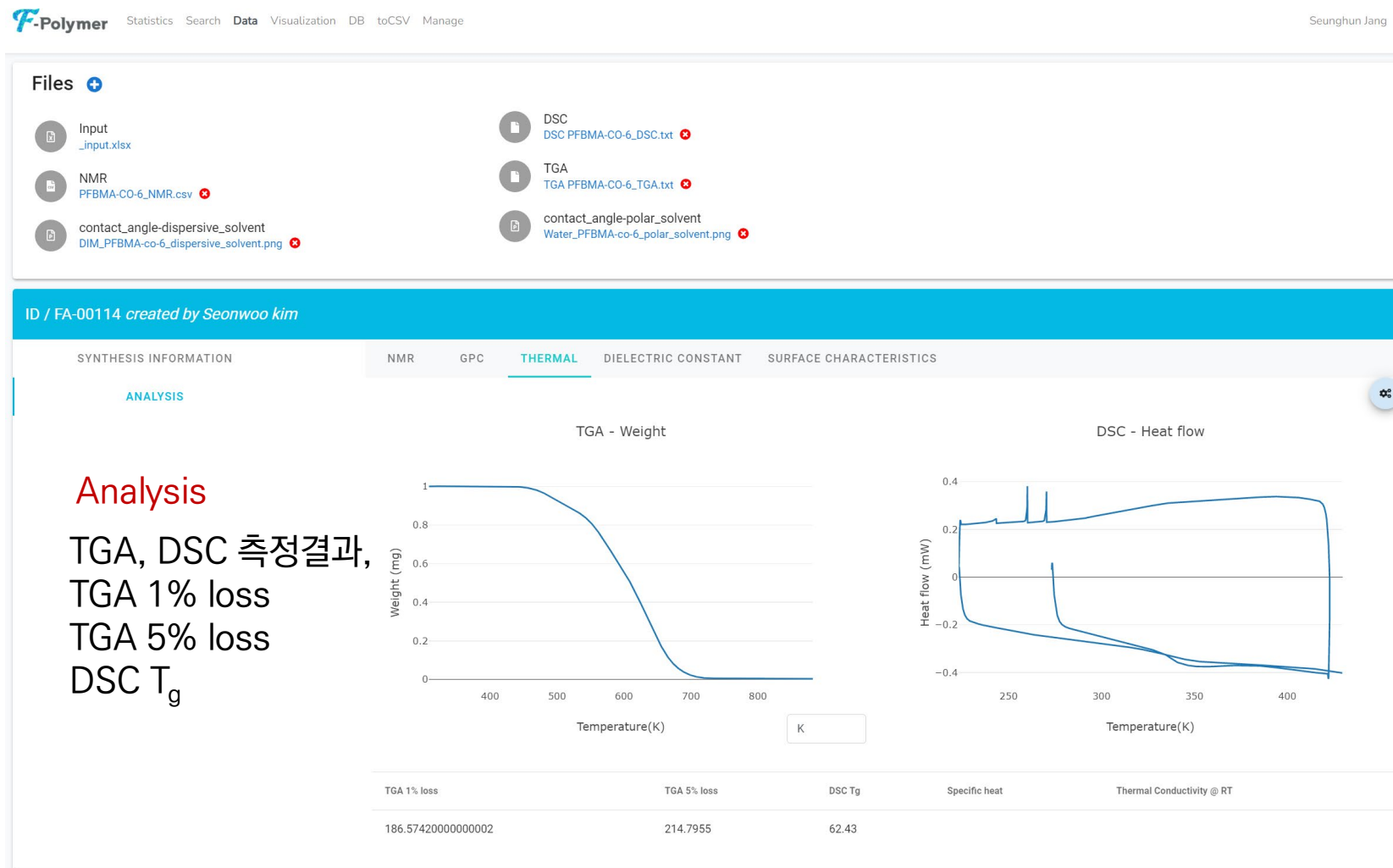
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Rows per page: 10

1-10 of 447

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Fluorinated Polymer Data platform: F-polymer



F-Polymer

f-polymer.chemdx.org

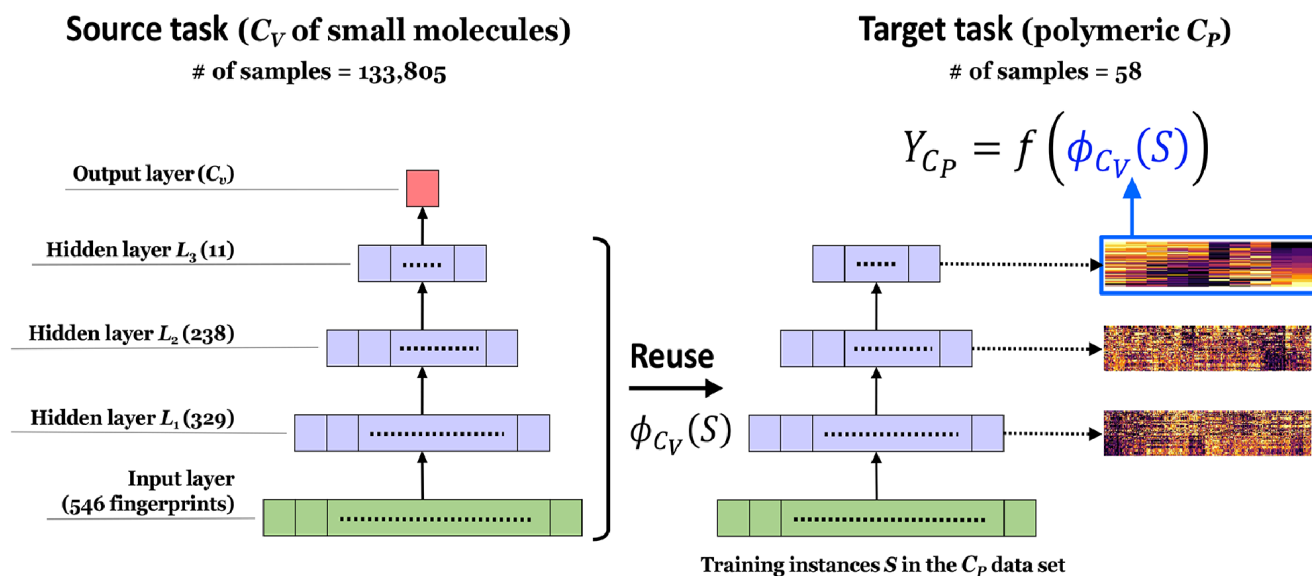
Fluorinated Polymer Data platform: F-polymer

- **F-Polymer** 플랫폼 내 API 활용 Dataset 다운로드.

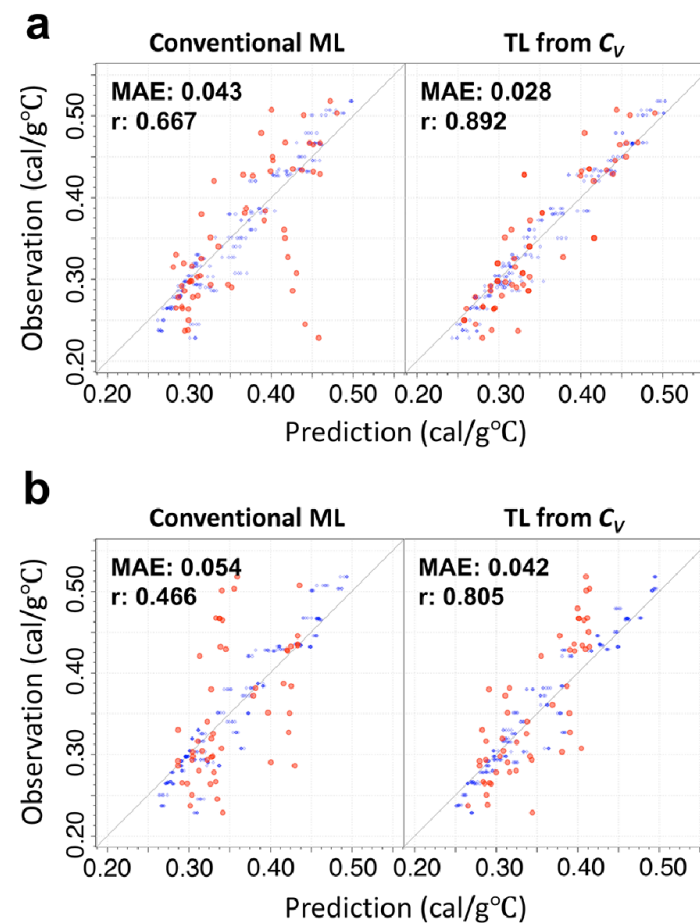
Surface En.
Diel Const.
DSC T_g

Exp_Name	CAS num.	Feed Ratio	SMILES																	
ID	Exp_Name	Monomer_a	Monomer_b	Monomer_c	Monomer_d	Monomer_e	Feed_ratio_a	Feed_ratio_b	Feed_ratio_c	Feed_ratio_d	Feed_ratio_e	Monomer_a	Monomer_b	Monomer_c	Monomer_d	Monomer_e	Surf_EN	1 kHz Diel. Const.	DSC T _g	
FA-00463	312,M2-E2-BU4-HMAM	80-62-6	97-63-2	97-88-1	79-41-4	115372-36-6	2	2	4	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	2.641264	71.97	
FA-00462	311,M2-E6-HMAM	80-62-6	97-63-2	97-88-1	79-41-4	115372-36-6	2	6	0	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	2.641264	92.21	
FA-00461	310,M6-E2-HMAM	80-62-6	97-63-2	97-88-1	79-41-4	115372-36-6	6	2	0	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	2.641185	116.11	
FA-00447	299,MMA-SMA-HMAM-442	80-62-6	32360-05-7	80-62-6	79-41-4	115372-36-6	4	4	0	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	2.2048	62.90	
FA-00414	271,HFBFA-SMA-HEMA-442	54052-90-3	32360-05-7	80-62-6	79-41-4	868-77-9	4	4	0	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	2.2160	59.8	
FA-00388	206,HFBFA-442	54052-90-3	32360-05-7	80-62-6	79-41-4	868-77-9	4	4	4	4	4	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	2.315714	50.0	
FA-00138	0027,HFBFA-CO-6	54052-90-3	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	0	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	14.26	2.05	36.45
FA-00137	0026,HFBFA-CO-4	54052-90-3	32360-05-7	80-62-6	79-41-4	868-77-9	4	1	2	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O		2.12	53.58
FA-00136	0025,HFBFA-CO-2	54052-90-3	32360-05-7	80-62-6	79-41-4	868-77-9	2	1	4	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	25.42		65.06
FA-00467	216,PF3-MMA-AA1	52591-27-2	32360-05-7	80-62-6	79-10-7	868-77-9	3	0	4	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.898288	64.70	
FA-00465	314,NFA3-MMA-AA1	52591-27-2	32360-05-7	80-62-6	79-10-7	868-77-9	3	0	4	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.341778	65.22	
FA-00464	313,NFA3-MMA-AA1	52591-27-2	32360-05-7	80-62-6	79-10-7	868-77-9	3	0	4	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.375707	64.36	
FA-00455	304,EMA-82	52591-27-2	97-63-2	80-62-6	79-10-7	868-77-9	0	8	0	0	0	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			80.04
FA-00451	303,NA51112_pMA	52591-27-2	32360-05-7	80-62-6	79-10-7	868-77-9	5	1	1	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	6.4	2.8745	1.21
FA-00436	388,NFA-SMA-HMAM-442	52591-27-2	32360-05-7	80-62-6	79-41-4	115372-36-6	4	4	0	0	0	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.2511	5.2	
FA-00413	270,NFA-SMA-HEMA-442	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	4	4	0	0	0	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	4.951	5.71	
FA-00391	200,NA304M12	52591-27-2	4813-57-4	80-62-6	79-10-7	818-61-1	3	0	4	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	7.01	2.252	52.3
FA-00390	199,NA30412	52591-27-2	4813-57-4	80-62-6	79-10-7	818-61-1	3	0	4	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	6.45	2.2785	1.00
FA-00389	198,NA31M312	52591-27-2	32360-05-7	96-33-3	79-10-7	818-61-1	3	1	3	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	10.52	2.2115	1.47
FA-00388	197,NA31312	52591-27-2	4813-57-4	96-33-3	79-10-7	818-61-1	3	1	3	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	10.7	2.2952	1.23
FA-00387	196,NA31112	52591-27-2	4813-57-4	96-33-3	79-10-7	818-61-1	5	1	1	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	6.3	3.5515	1.82
FA-00385	194,NFA-604	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	1	4	4	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.25	7.735	52.7
FA-00384	193,NFA-613	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	1	3	3	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.16	5.6042	56.2
FA-00383	192,NFA-622	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	2	2	2	2	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.84	5.5050	50.8
FA-00382	191,NFA-631	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	3	1	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.12	2.5056	13.12
FA-00381	190,NFA-640	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	4	4	4	4	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	8.25	2.5112	0.1
FA-00341	143,NFA-442	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	4	4	4	4	4	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O		3.38866	46.2
FA-00249	0135,NFA-60-10-0-20-10	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	0	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	8.2	2.4162	65.0
FA-00248	0134,NFA-60-10-0-15-15	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	0	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	7.2	2.1155	72.01
FA-00247	0133,NFA-60-10-0-5-25	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	0	0.5	0.5	2.5 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	11.2	2.2152	83.45
FA-00246	0132,NFA-60-10-0-0-30	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	0	0	0	3 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	14.7	2.1962	29.1
FA-00208	0104,NFA-CO-nmoH	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	2	1	1	0 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.9	2.2005	29.22
FA-00207	0103,NFA-CO-nmoH	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	1	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	10.7	1.1355	29.22
FA-00135	0024,NFA-CO-6	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	6	1	0	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	4.9	2.11	32.55
FA-00134	0023,NFA-CO-4	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	4	1	2	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			52.59
FA-00133	0022,NFA-CO-2	52591-27-2	32360-05-7	80-62-6	79-41-4	868-77-9	2	1	4	1	1	2 F(C)F(C)F(C)F(C)F(C)C(C)=O	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	15.4		2.06
FA-00297	87,PF3-SUMA-CO-1	45115-53-5	38862-25-8	80-62-6	79-41-4	868-77-9	2	1	4	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O		1.9814612	132.4
FA-00452	85,PF3-SUMA-CO-5	45115-53-5	38862-25-8	80-62-6	79-41-4	868-77-9	2	5	0	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O	5.588		150.46
FA-00296	78,PF3-PAA-CO-1	45115-53-5	1611-83-2	80-62-6	79-41-4	868-77-9	2	1	4	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O		1.049084	111.54
FA-00295	77,PF3-PAA-CO-3	45115-53-5	1611-83-2	80-62-6	79-41-4	868-77-9	2	3	2	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			69.81
FA-00294	76,PF3-PAA-CO-5	45115-53-5	1611-83-2	80-62-6	79-41-4	868-77-9	2	5	0	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			67.2
FA-00466	315,PF3-MMA-MMA1	45115-53-5	32360-05-7	80-62-6	79-41-4	868-77-9	3	0	4	1	1	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			98.88
FA-00460	309,PF1-M7-HMAM	45115-53-5	52591-27-2	80-62-6	79-41-4	115372-36-6	1	0	7	0	7	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			102.27
FA-00459	308,PF3-M5-HMAM	45115-53-5	52591-27-2	80-62-6	79-41-4	115372-36-6	3	0	5	0	5	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			103.21
FA-00458	307,PF2-M6-HMAM	45115-53-5	52591-27-2	80-62-6	79-41-4	115372-36-6	2	0	6	0	6	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			115.69
FA-00448	300,MMA-SMA-HMAM-262	45115-53-5	32360-05-7	80-62-6	79-41-4	115372-36-6	6	2	0	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C	CCCCOC(=O)C(C)=C	CC(=C)C(=O)O	CC(=C)C(=O)O	CC(=C)C(=O)O			2.204
FA-00417	274,PPMA-SMA-HEMA-442	45115-53-5	32360-05-7	80-62-6	79-41-4	868-77-9	4	4	0	0	0	2 C(C)=O(C)C(C)=C	CCCCOC(=O)C(C)=C							

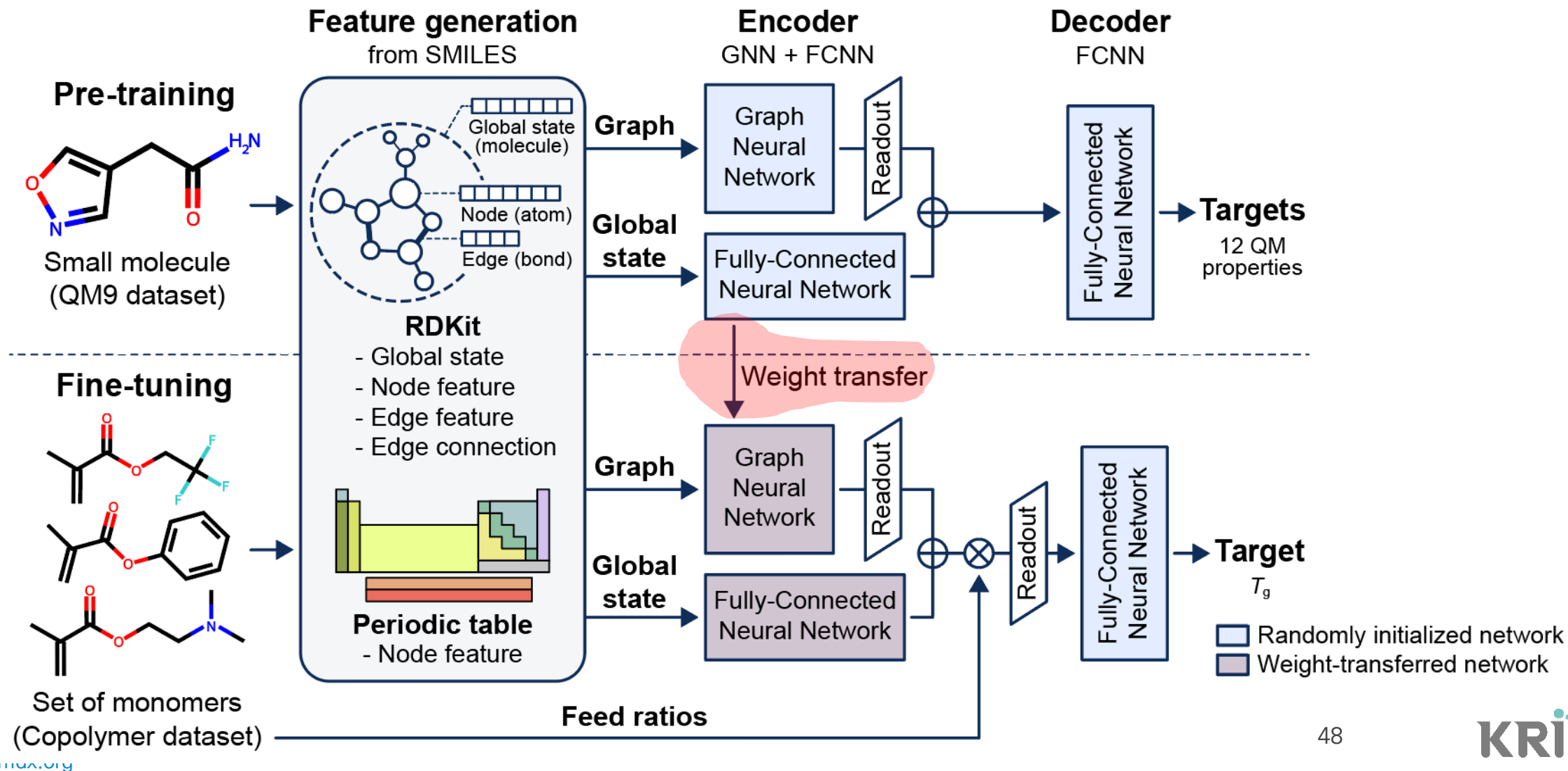
Transfer learning



ACS Cent. Sci. 2019, 5, 1717–1730



Model training procedure



Pre-training results

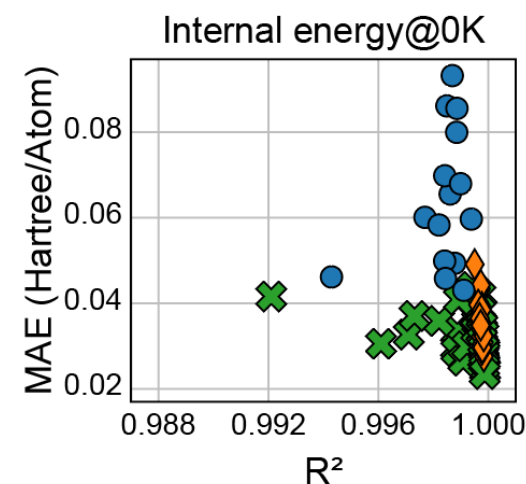
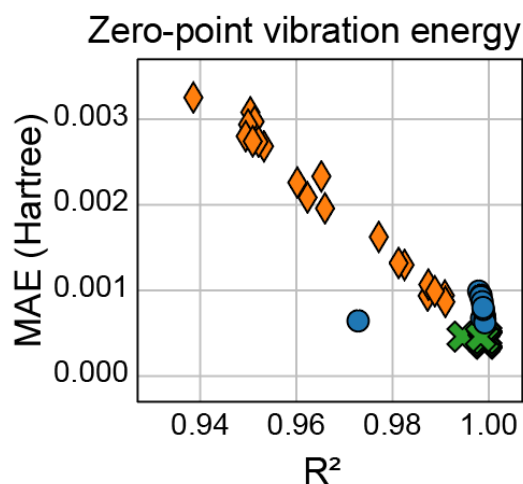
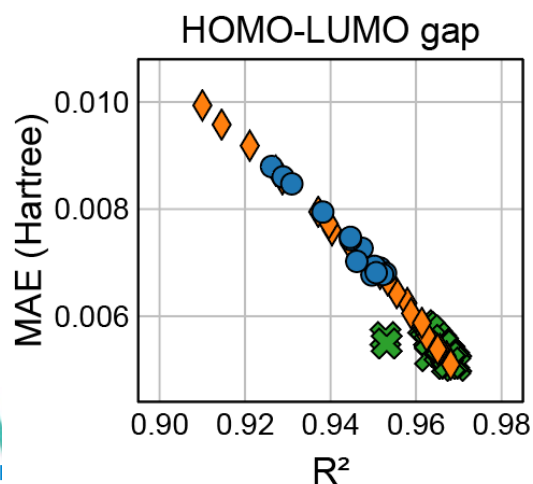
- Hyperparameter optimization (model parameters)

Network Type	Feature Arch.		GNN Nodes	Layers	FCNN Nodes	Layers	Total
Encoder	GNN	◆ Graph	8, 16, 32, 64, 128	2, 4, 6, 8	–	–	20
	FCNN	● Global state	–	–	32, 64, 128, 256, 512	1, 2, 3	15
	Hybrid	✕ Graph + Global state	32, 64, 128	4, 6, (8) ^{a)}	128, 256	1, 2, (3) ^{b)}	40
Decoder	FCNN	–	–	–	128	2	1

^{a)}GNN with 8 layers for the hybrid encoder is used for 32 and 64 nodes.

^{b)}FCNN with 3 layers for the hybrid encoder is used for 128 nodes.

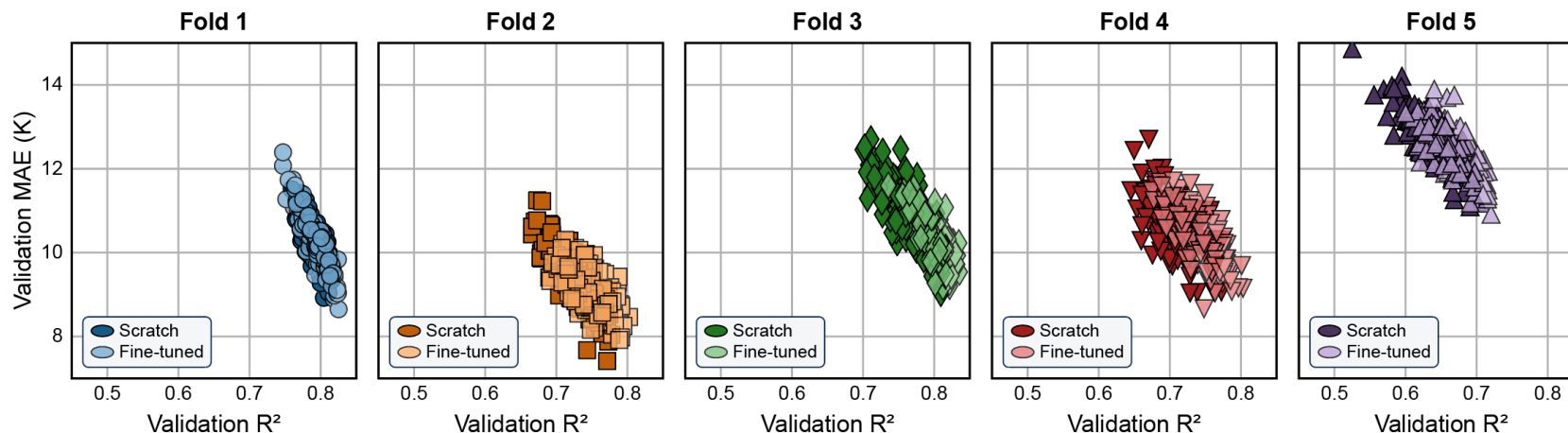
- Example outputs (average of validation performance from 5-fold CV)



Model parameters exhibit the highest R^2 and the lowest MAE across all 12 targets are selected

Fine-tuning results (copolymer dataset)

- Model performances (5-fold CV / 144 models from H.P. opt.)



Model	Feature	R ²	MAE (K)
XGB	Morgan FP	0.708 ± 0.092	11.602 ± 1.208
	MACCS key	0.720 ± 0.046	11.377 ± 1.161
SVR	Morgan FP	0.682 ± 0.077	12.308 ± 1.957
	MACCS key	0.708 ± 0.075	11.775 ± 1.112
Scratch	Graph	0.752 ± 0.052	10.619 ± 1.306
	Global state	0.745 ± 0.059	10.506 ± 1.107
	Hybrid	0.742 ± 0.045	10.758 ± 0.979
Fine-tuned	Graph	0.779 ± 0.035	10.018 ± 1.132
	Global state	0.736 ± 0.055	10.541 ± 0.927
	Hybrid	0.773 ± 0.040	10.022 ± 0.819

Data scarcity

- Limited data sampling
- Uneven data distribution & uneven data split
- Performance variability (high STD)
- Low reliability of predictions

Ensemble model

• Ensemble strategy

Methods

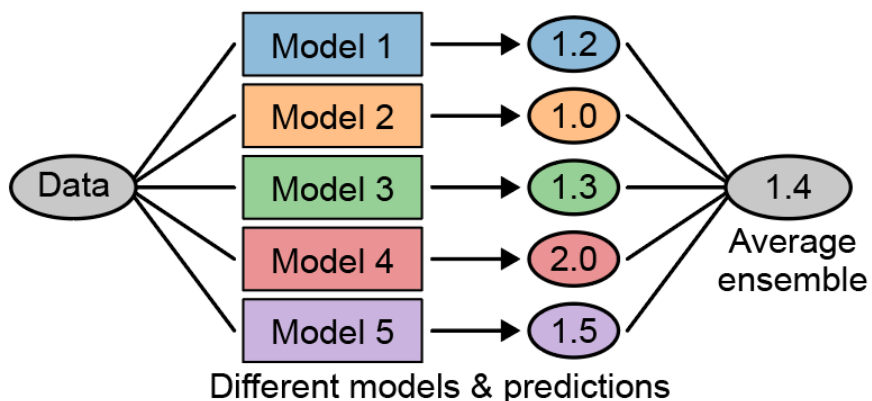
- Bagging
- Boosting
- Stacking
- Negative correlation learning
- Explicit / Implicit
- Homogeneous / Heterogeneous

Fusion strategy

- Unweighted model averaging
- Majority voting
- Bayes optimal classifier
- Stacked generalization
- Super learner
- Consensus
- Query-by-committee

• Ensemble of regression models

(stacking / explicit / homogeneous / unweighted)

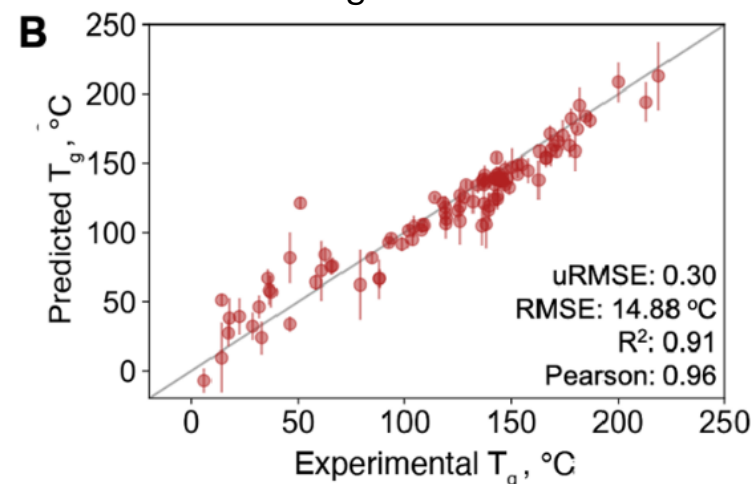


- More robust and accurate results
- No extra meta-learner (small dataset)

Highest R^2 models validation / different hyperparameters

Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
R^2	0.817	0.794	0.821	0.797	0.720
MAE (K)	10.10	8.79	9.29	9.73	10.92

• Polymers' T_g prediction

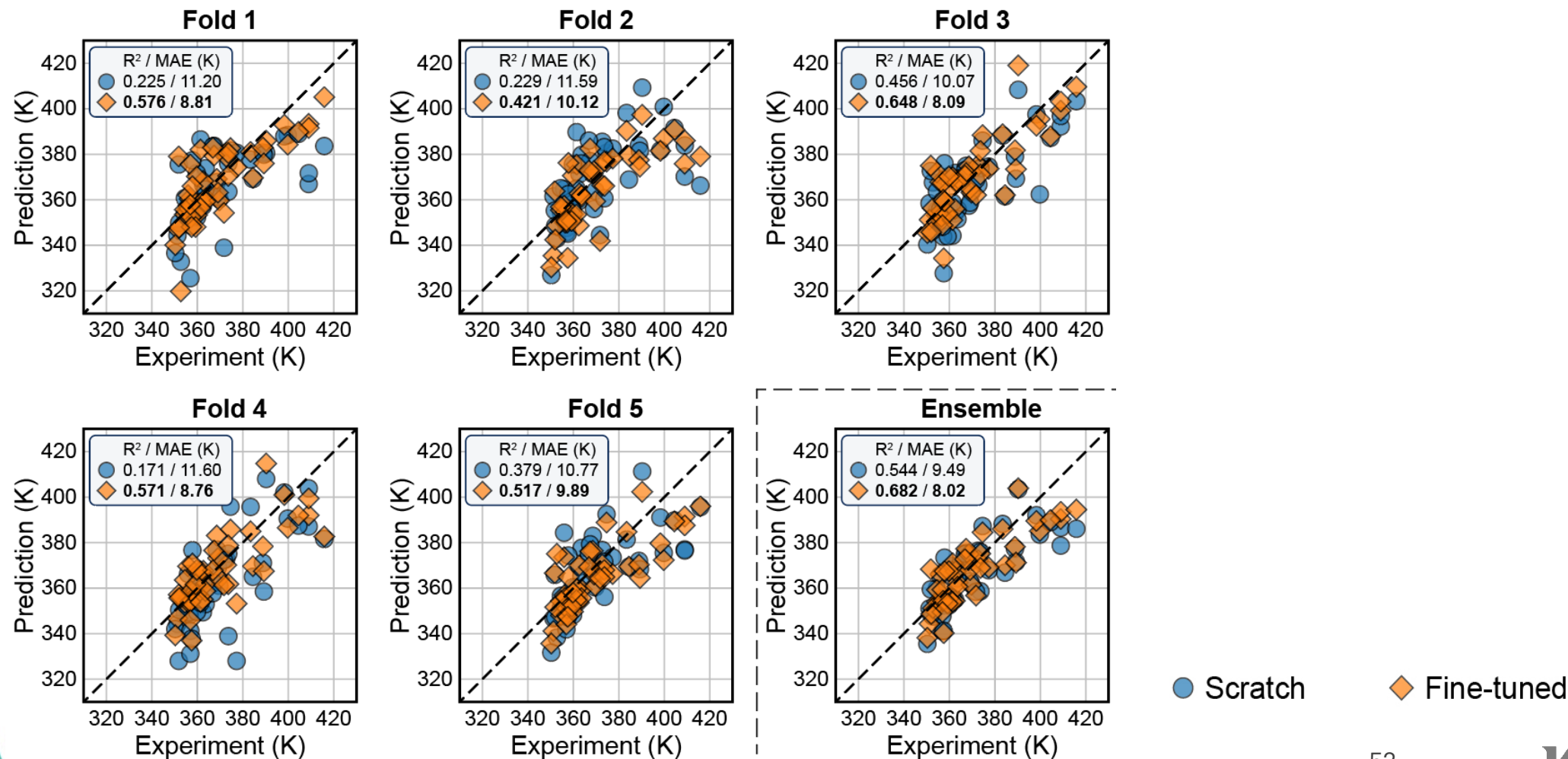


M.Ganaie *et al.*, *Eng. Appl. Artif. Intell.* 115 (2022) 105151

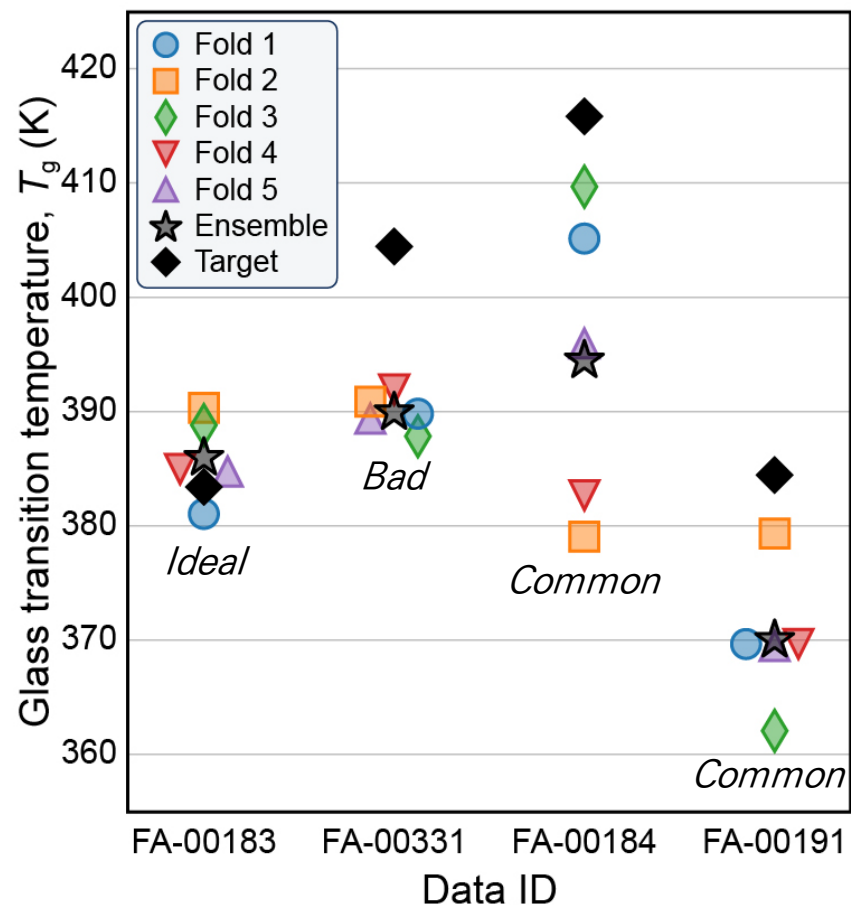
Y.S.AIFaraj *et al.*, *ACS Cent. Sci.* 9 (2023) 1810

Ensemble model performance (test dataset)

Average ensemble of the *highest validation R^2 models* from *each fold*.



Prediction examples for test data



FA-00183 – Ideal case

- High accuracy with low variation between predictions from each model.

FA-00331 – Bad case

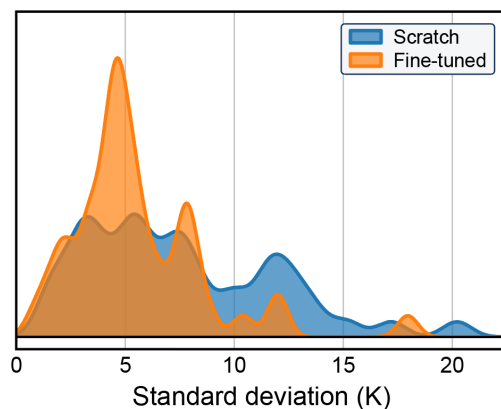
- Low accuracy with low variation between predictions from each model.
- Similar inaccurate predictions due to data scarcity.

FA-00184 & FA-00191 – Common cases

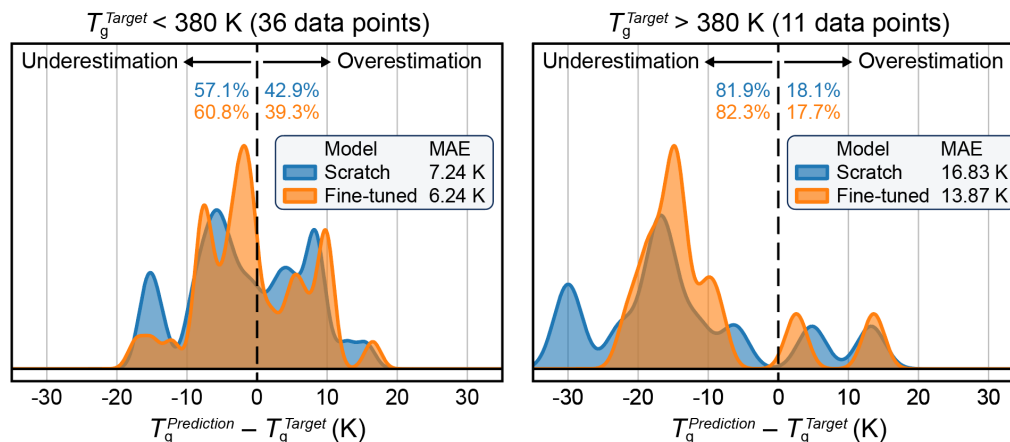
- Noticeable variation among predictions.
- No single model uniformly outperformed. (Orange – Fold 2 Vs. Green – Fold 3)
- Inconsistent model performance due to uneven data distributions.

Effect of fine-tuning – Test dataset

Standard deviation



Error distributions



Distribution of standard deviations (STDs)

Scratch: no peak, distributed up to 20 K

Fine-tuned: peak near 5 K, distributed up to 13 K

→ Lower uncertainty of the fine-tuned model.

Error distributions for $T_g < 380$ K

Low MAEs with normal-like distributions for both scratch and fine-tuned models.

Error distributions for $T_g > 380$ K

Underestimation of prediction values due to biased training data.

High MAEs of 16.83 K and 13.87 K for scratch and fine-tuned models, respectively.

→ Decreased underestimation trends and MAE for fine-tuned model.

Hypothetical monomer combinations

- Data generation

$$*f_a + f_b + f_c + f_d = 0.8$$

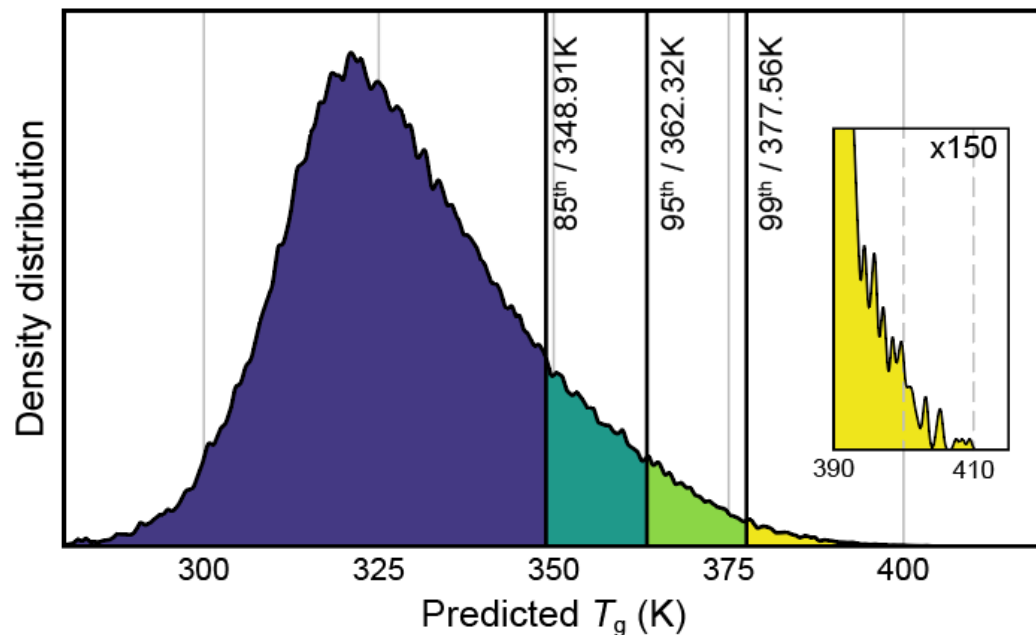
Monomer		Feed ratio
Monomer A	One of fluoride (22 monomers) One of non-fluoride (except Monomer in "E", 36 monomers)	f_a : 0.2 / 0.4 / 0.6
Monomer B		f_b : 0.0 / 0.2 / 0.4 / 0.6
Monomer C		f_c : 0.0 / 0.2 / 0.4 / 0.6
Monomer D		f_d : 0.0 / 0.2 / 0.4 / 0.6
Monomer E	Hydroxyethyl methacrylate (fixed)	0.2 (fixed)

- From a set of 22 fluorinated monomers, one was selected with feed ratios of 0.2, 0.4, 0.6, and 0.8.
- The monomer hydroxyethyl methacrylate (HEMA) was consistently used with a feed ratio of 0.2 to facilitate polymer film formation.
- The remaining three monomers were selected from 36 non-fluorinated monomers without duplication, with possible feed ratios of 0, 0.2, 0.4, and 0.6.
- Sum of feed ratios must be 1.

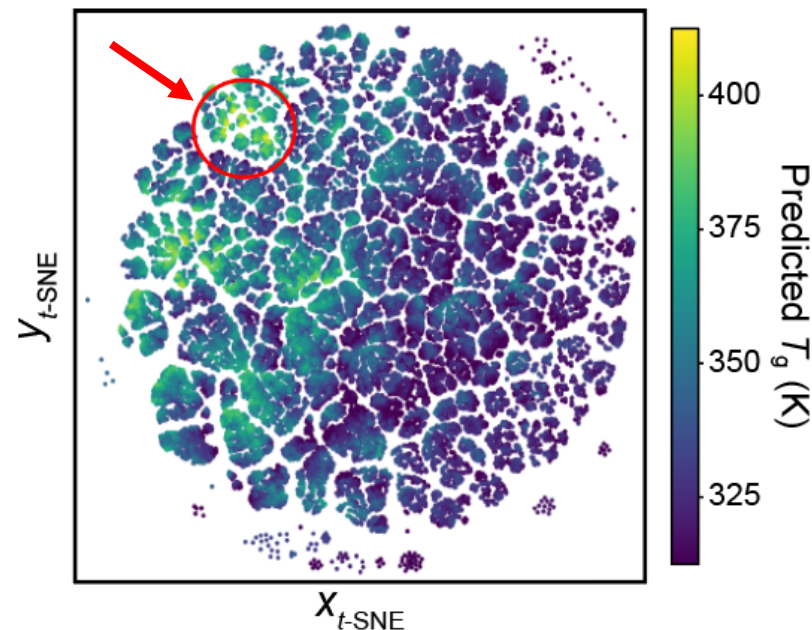
Total 201,850 monomer combinations

Prediction results using fine-tuned model

- Distribution of T_g for 201,850 datapoints

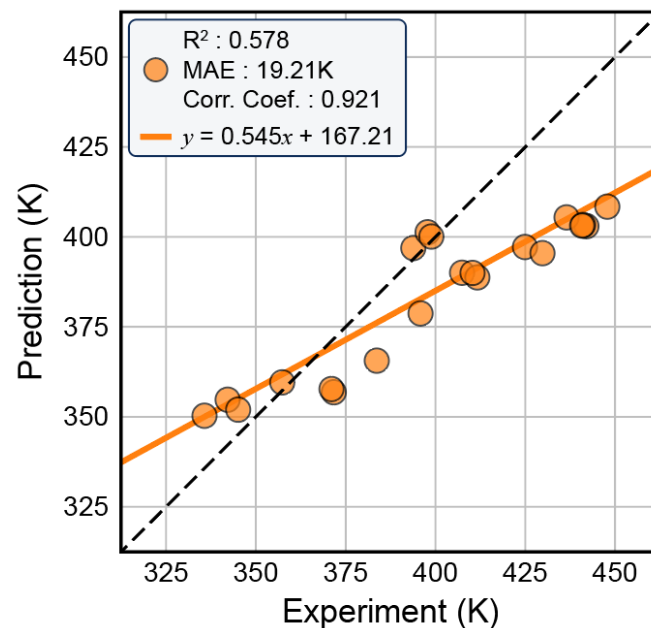
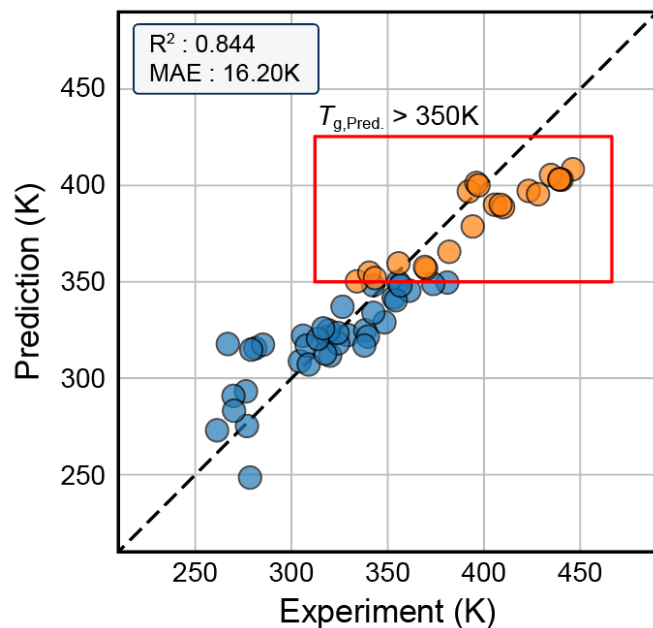


- The distribution peaks around 321 K and gradually declines.
- About 85% of data have T_g values lower than 350 K. (lower boundary of test data)
- 0.14% of data (274 entries) exceeds 390 K and 0.01% of data (28 entries) exceeds 400 K. (constraints from training dataset)



- High T_g datapoints are clustered within t -SNE space.
- There is shared chemical information for high T_g datapoints.

Experimental validation



- All entries (blue and orange)
 - High R^2 of 0.844 for 58 entries
- High T_g entries (predicted T_g over 350K, orange)
 - Low R^2 of 0.578 for 21 entries due to underestimation of T_g values. (bias from training dataset)
 - High correlation coefficient of 0.921 between experiment and prediction values.
 - High T_g over 400K of 10 new entries.

Experimental validation results ($T_{g, \text{Pred.}} > 390\text{K}$)

No.	Monomers					Feed ratios					Pred. (K)	Exp. (K)
	A	B	C	D	E	A	B	C	D	E		
1	HFIPMA ^{a)}	IBMA ^{e)}	NSMA ^{g)}	–	HEMA	0.2	0.4	0.2	–	0.2	390.03	405.68
2	TFPMA ^{b)}	IBMA	NSMA	MAA ^{k)}	HEMA	0.2	0.2	0.2	0.2	0.2	395.48	428.08
3	TFEMA ^{c)}	IBMA	NSMA	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	396.90	392.10
4	PFPMA ^{d)}	IBMA	NSMA	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	397.19	423.20
5	HFIPMA	ADMMA ^{f)}	–	MAA	HEMA	0.4	0.2	–	0.2	0.2	400.10	397.13
6	HFIPMA	ADMMA	MMA ^{h)}	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	401.28	396.07
7	TFEMA	ADMMA	NSMA	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	402.98	440.32
8	HFIPMA	ADMMA	IBMA	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	403.14	439.12
9	HFIPMA	ADMMA	DCPMA ⁱ⁾	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	403.31	439.40
10	HFIPMA	ADMMA	MMAA ^{j)}	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	405.42	434.78
11	HFIPMA	ADMMA	NSMA	MAA	HEMA	0.2	0.2	0.2	0.2	0.2	408.41	446.14

a) Hexafluoroisopropyl methacrylate

b) Tetrafluoropropyl methacrylate

c) Trifluoroethyl methacrylate

d) Pentafluoropropyl methacrylate

e) Isobornyl methacrylate

f) Adamantyl methacrylate

g) N-succinimidyl methacrylate

h) Methyl methacrylate

i) Dicyclopentanyl Methacrylate

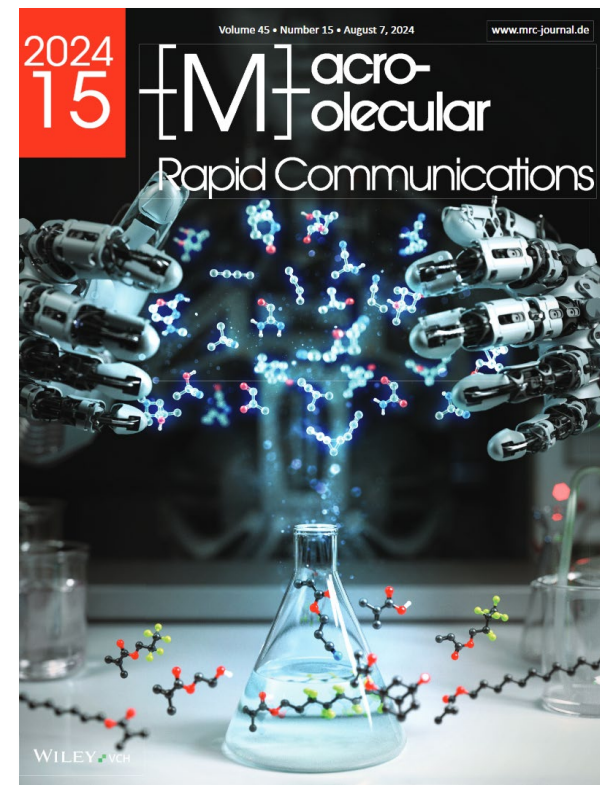
j) N-Methylmethacrylamide

k) Methacrylic acid

Monomer	# of entries ($T_g > 390\text{K}$)	# of entries (Train/Valid)	T_g range (K) (Train/Valid)	High T_g s (K) (Train/Valid)
HFIPMA ^{a)}	57 (20.8%)	12	306 ~ 416	415.58 / 406.77 / 380.43 / 360.79 / ...
TFEMA ^{c)}	34 (12.4%)	7	305 ~ 361	361.18 / 357.92 / 357.55 / 354.64 / ...
IBMA ^{e)}	65 (23.7%)	0	–	–
NSMA ^{g)}	94 (34.3%)	2	–	405.55 / 373.61

Summary

- Designing high T_g fluorinated polymers using machine learning.
- Limited fluorinated polymer dataset
 - Knowledge-based initiation – limited by the researcher's experience.
 - Limited amount – biased within chemical space.
- *Machine learning modeling*
 - Feature generation using RDKit for small organic molecules.
 - Transfer learning using the QM9 dataset (similar atomistic information) for improved accuracy.
 - Ensemble modeling for enhancing prediction accuracy and reliability.
- *Screening and validation*
 - Screening more than 200k hypothetical monomer combinations using the fine-tuned ensemble model.
 - Discovery of new high T_g fluorinated polymers via experimental validation. (10 polymers with $T_g > 400$ K and 5 polymers with $T_g > 430$ K).



Macromol. Rapid Commun. 2024 45 2400161

데이터셋, 데이터 수집 및 인공지능 플랫폼

무기 형광체 데이터셋

- <https://doi.org/10.6084/m9.figshare.24771186>
- <https://github.com/KRICT-DATA/IPOP-dataset-ver-3.0>
- S Jang*, GS Na, Y Choi, H Chang, "Optical property dataset of inorganic phosphor" Sci. reports (2024) 14, 7639

분자쌍 계산 데이터셋 및 데이터 생성 파이썬 코드

- <https://doi.org/10.6084/m9.figshare.29105159> (데이터셋)
- <https://github.com/SeunghunJang/MPSGO> (파이썬 코드)

실험데이터 수집 플랫폼 (일부 데이터 공개)

- polymer.chemdx.org (고분자 소재)
- f-polymer.chemdx.org (불소고분자 소재)
- sensor.chemdx.org (센서 소재)

인공지능 플랫폼

- simpl-ml.org (머신러닝 알고리즘 학습 가능)
- ai.chemdx.org (분자 및 결정구조 기반 그래프 인공지능, 이미지 학습 가능)

Thank you for your attention.
Please feel free to ask any questions.